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Beta-Glucan Modulates Monocyte Plasticity and Differentiation Capacity to Mitigate DSS-Induced Colitis

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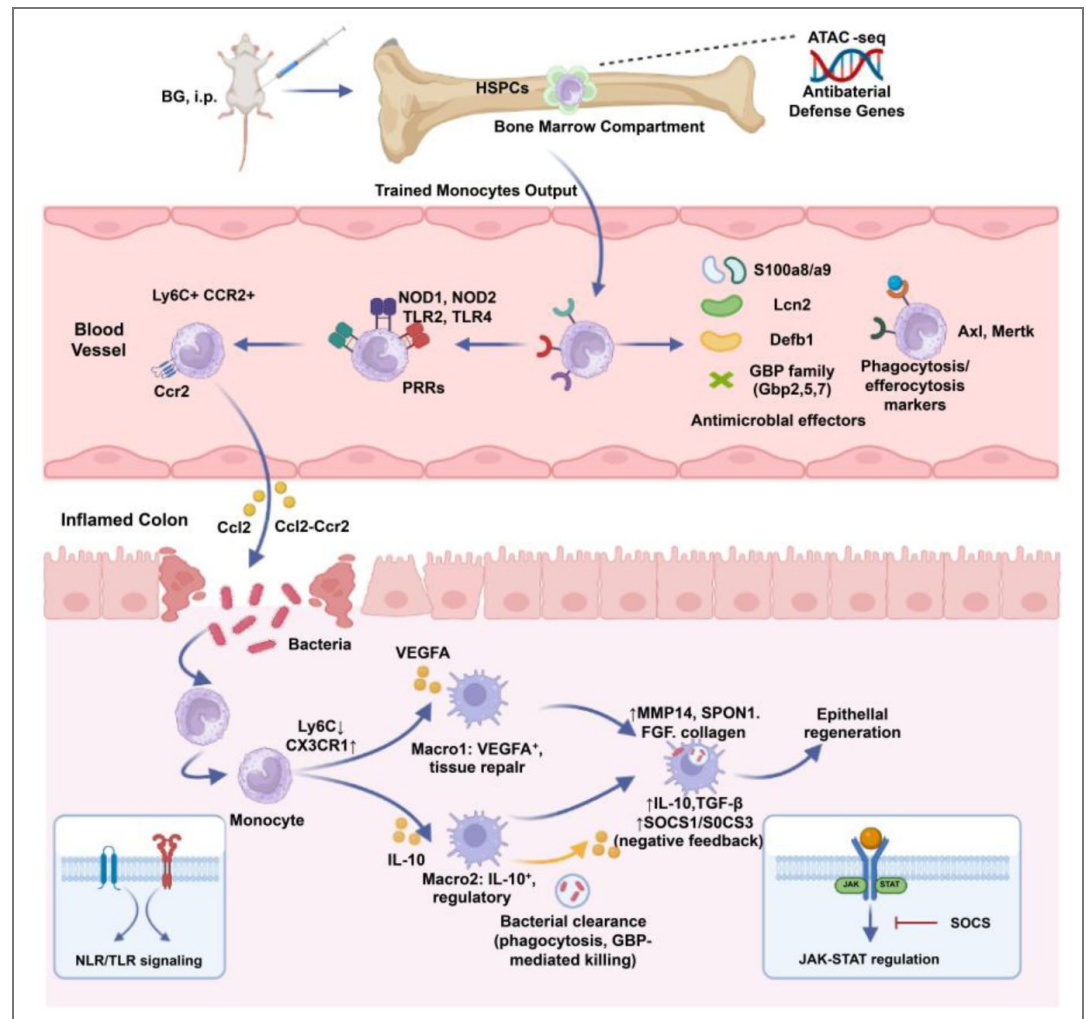
This study presents **compelling** evidence supporting the therapeutic potential of trained immunity in Colitis. The study is **important** for the field of trained immunity and is a welcome addition to the focus issue on trained immunity.

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Abstract

Trained immunity involves the reprogramming of innate immune cells after an initial exposure, resulting in heightened inflammatory responses to subsequent stimuli and enhanced bactericidal capacity during infection. However, this proinflammatory state could also exacerbate chronic conditions like inflammatory bowel disease (IBD), which is characterized by persistent inflammation and microbial imbalance. It remains unclear how trained immunity influences IBD pathogenesis and whether it can be harnessed therapeutically. In our study, pretreatment with β -glucan reprogrammed bone marrow hematopoietic progenitors and peripheral monocytes, inducing a profound shift in monocyte plasticity and significantly reducing the severity of dextran sulfate sodium (DSS)-induced colitis. Adoptive transfer of bone marrow or peripheral monocytes from β -glucan-trained mice into naive mice conferred robust protection against colitis, demonstrating that this protective effect is transferable. Trained mice also displayed improved clearance of intestinal bacterial infections. Single-cell RNA sequencing revealed an expansion of reparative Cx3cr1⁺ macrophages derived from Ly6C^{hi} monocytes, correlating with accelerated colonic epithelial regeneration. Collectively, these findings reveal how β -glucan-induced trained immunity modulates monocyte differentiation to ameliorate experimental colitis, highlighting the potential of harnessing trained immunity as a therapeutic strategy to recalibrate innate immune responses and restore gut homeostasis in IBD, shedding light for future clinical applications.

Graphical abstract



Introduction

Inflammatory bowel disease (IBD), encompassing ulcerative colitis (UC) and Crohn's disease (CD), is a chronic gastrointestinal inflammatory disorder associated with significant morbidity^(1, 2). The intestinal mucosa is continuously exposed to a diverse microbial ecosystem, including fungi, bacteria, and viruses⁽³⁾. In IBD, the delicate balance between the immune system and the microbiota is disrupted, leading to uncontrolled inflammation^(4, 5). This dysregulation is further exacerbated by genetic factors, such as loss-of-function mutations in NOD2, which impair antimicrobial peptide production, increase susceptibility to intestinal infections, and decrease IL-10 production by monocytes^(6–8).

Trained immunity, a phenomenon where innate immune cells exhibit enhanced responses to secondary challenges, has emerged as a promising therapeutic avenue for various infectious diseases and cancer^(9–13). However, its role in chronic inflammatory conditions like IBD remains largely unexplored. While trained immunity can enhance pathogen clearance, it also involves the upregulation of pro-inflammatory cytokines such as TNF, IL-1 β , and IL-6⁽¹⁴⁾, therefore the role of trained immunity in IBD remained elusive.

This study aimed to investigate the therapeutic potential of trained immunity in a mouse model of colitis. We hypothesized that inducing trained immunity could enhance microbial control, thereby preserving intestinal integrity and ultimately mitigating disease severity. To test this, we employed β -glucan (BG), a prototypic trained immunity inducer, and assessed its impact on dextran sulfate

sodium (DSS)-induced colitis. Our findings demonstrate that BG pretreatment significantly alleviates colitis in mice. Mechanistically, BG induced central immunity in the hematopoietic compartment, leading to the generation of “trained” monocytes in the periphery. Both bone marrow transplantation from trained donors and adoptive transfer of trained monocytes conferred protection against DSS-induced colitis. Furthermore, single-cell RNA-seq revealed that BG-induced trained immunity promotes the expansion of reparative Cx3cr1 intestinal macrophages derived from Ly6C^{hi} monocytes, thereby facilitating epithelial regeneration. In summary, our data suggest that harnessing trained immunity represents a promising therapeutic approach for IBD treatment.

Results

β -glucan (BG) pretreatment ameliorates DSS-induced colitis

To investigate the impact of trained immunity on colitis, mice were pretreated with BG, a potent inducer of trained immunity. Successful induction of BG-induced trained immunity (BGTI) was confirmed by the enhanced resistance to *Staphylococcus aureus* infection in BG-treated mice (Fig S1A [↗](#) and S1B [↗](#)), consistent with previous findings(15). Importantly, BG-pretreatment did not induce spontaneous intestinal inflammation (Fig S1C [↗](#) and S1D [↗](#)). Following BGTI induction, mice were subjected to DSS-induced colitis. (Fig 1A [↗](#)). BG pretreatment significantly ameliorated disease severity, as indicated by the reduced body weight loss (Fig 1B [↗](#) and Fig S1E [↗](#)), colon shortening (Fig 1C [↗](#) and Fig S1F [↗](#) and S1G [↗](#)), and diminished mucosal inflammation observed via endoscopic imaging (Fig 1D [↗](#)). Histological analysis demonstrated lower histopathologic scores (Fig 1E [↗](#) and 1F [↗](#)). Additionally, BG-pretreated mice exhibited enhanced intestinal epithelial barrier integrity, as evidenced by the increased expression of tight junction proteins Zo-1 and Occludin (Fig 1G [↗](#) and 1H [↗](#)), and decreased circulating FITC-dextran levels (Fig 1I [↗](#)).

Acknowledging the long-lasting effects of BGTI(16), we further examined its long-term protective effects against DSS-induced colitis. Despite peripheral myeloid cells and monocytes returning to homeostasis four weeks post-BG treatment (Fig S2A [↗](#) and S2B [↗](#)), mice remained protected against DSS-induced colitis as demonstrated by body weight change and colon shortening (Fig 1J [↗](#) and 1K [↗](#)), and improved histological outcomes (Fig S2C [↗](#) and S2D [↗](#)). This BG-induced protection persisted for up to seven weeks (Fig S2E [↗](#) and S2F [↗](#)). These findings indicate that BG pretreatment effectively ameliorates DSS-induced colitis in mice, emphasizing the therapeutic potential of trained immunity in IBD management.

BG ameliorates colitis via enhanced myeloid cell activation independent of adaptive immunity

Given that BGTI has been reported to function independently of adaptive immunity(17, 18), we evaluated the effects of BG pretreatment in *Rag1* knockout mice (Fig S3A [↗](#)-S3D [↗](#)). Notably, BG pre-treatment remained effective in *Rag1*^{-/-} mice, as evidenced by reduced body weight loss, colon shortening, and improved histological outcomes (Fig S3E [↗](#)-S3H [↗](#)). These findings confirm that BGTI-mediated protection is independent of adaptive immune responses, consistent with previous findings demonstrating that adaptive immunity is dispensable for trained immunity.

To gain a comprehensive understanding of the molecular mechanisms underlying BG-mediated protection, we performed RNA sequencing on colonic samples collected at multiple time points following DSS administration. WGCNA identified the METurquoise module emerged as the core functional module, containing 5,015 genes whose expression exhibited significant co-regulation following BG intervention (Fig S4A [↗](#)). And differential gene expression of genes within this module was most significant on day 7 post-DSS treatment when comparing the BG-treated mice to PBS controls (Fig S4B [↗](#)). GO analysis revealed significant enrichment in pathways related to chemotaxis and leukocyte migration (Fig 2A [↗](#)). KEGG pathway enrichment analysis demonstrated significant activation of cellular processes involved in phagocytic function (Fig 2B [↗](#)). Additionally, the enrichment observed in WGCNA was further corroborated by whole-genome analysis of day 7 colitis samples, which also identified significant enrichment of pathways associated with

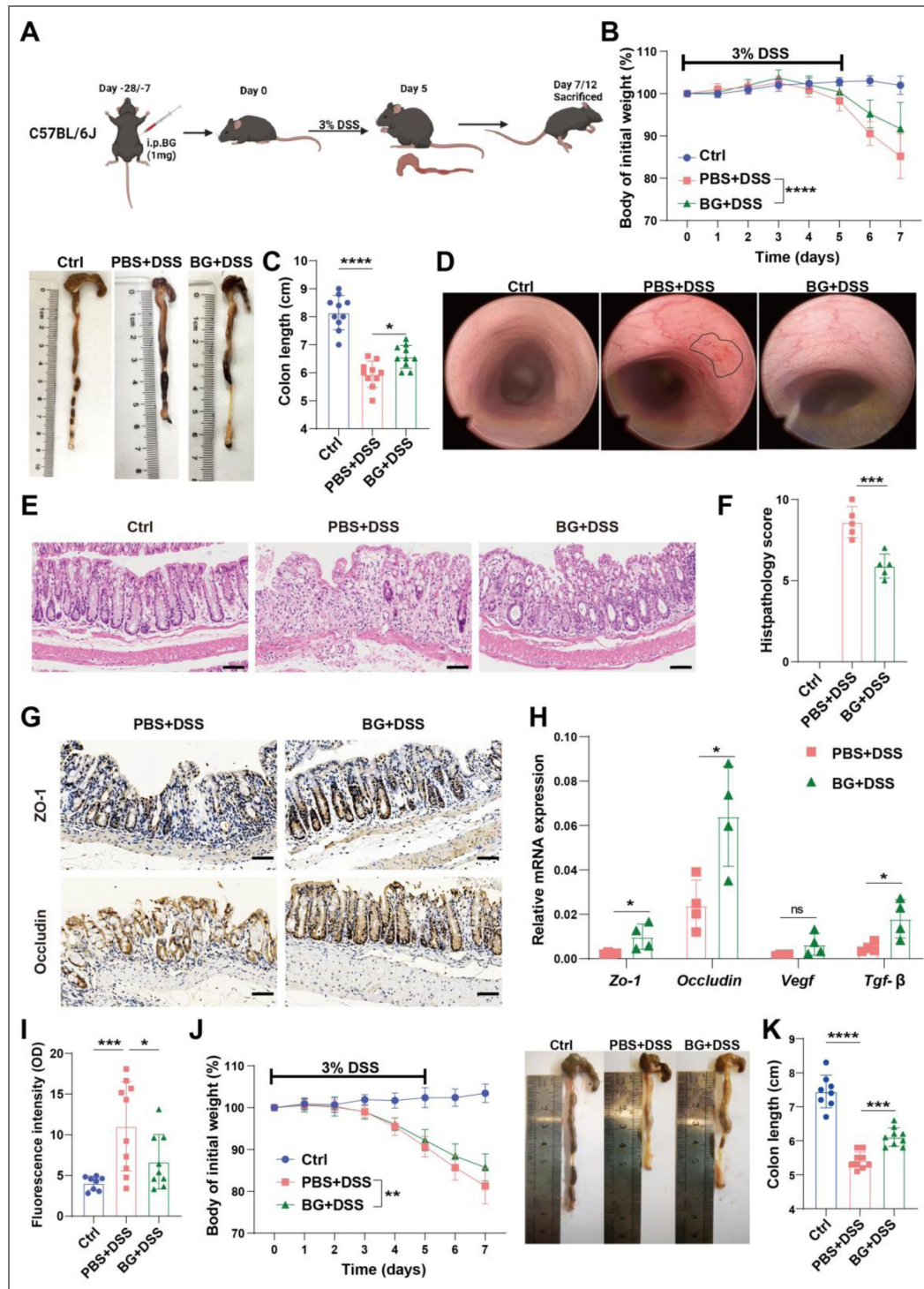


Fig 1. β-glucan (BG) pretreatment ameliorates DSS-induced colitis.

(A) Schematic representation of BG-induced trained immunity and DSS colitis model, (B) Body weight change curve of mice pretreated with BG for one week, followed by colitis induction with 3% DSS ($n = 10$), (C) Colon length changes in mice from (B), (D) Endoscopic images displaying mucosal damage, (E and F) H&E staining and histological scoring. Scale bars: 100 μm, (G and H) Expression levels of tight junction and repair, (I) FITC-dextran assay assessing intestinal barrier function, (J) Body weight change curve of mice pretreated with BG for four weeks, followed by colitis induction with 3% DSS ($n = 8-9$), (K) Colon length changes in mice from (J). Data are presented as mean $\pm$ SD. Statistical significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. ns, not significant.

chemotaxis, leukocyte migration and phagocytic function (Fig S4C [↗](#) and S4D [↗](#)). Multiple analytical approaches concurrently indicated that BG pretreatment not only enhances the recruitment of immune cells but also promotes their functional activation, particularly with respect to phagocytic capabilities.

To further investigate how BG pretreatment reprograms of intestinal myeloid cells, we performed single-cell RNA-seq to characterize intestinal leukocytes populations on day 7 post-DSS treatment. Unbiased clustering analysis identified multiple clusters of intestinal immune cells subsets, including monocytes/macrophages (*Ly6c2*, *Ccr2* and *Adgre1*), dendritic (DC) (*Cst3* and *H2-Aa*), neutrophils (*S100a8/a9* and *Ly6g*), B (*Cd79a* *Cd79b* and *Cd19*), Pre B (*Myl4* and *Mme*), CD4 T (*Cd3d* and *Cd4*), CD8 T (*Cd3d* and *Cd8a*), natural killer (*Nkg7*), ILC2 (*Gata3* and *Il4*) and ILC3 (*Rorc* and *Il22*) cells (Fig 2C [↗](#) and Fig S4E [↗](#)). Importantly, the ratio of monocytes/macrophages and neutrophils was elevated in the BG group (Fig 2D [↗](#)).

Single-cell RNA-sequencing data further demonstrated enhanced expression of innate pathogen recognition receptors (*Syk*, *Nod1*, *Nod2*, *Tlr2* and *Tlr4*), efferocytosis (*Cd300lf*, *Axl*, *Mertk*, *Itgal* and *Itgb7*), cytokines (*Il1b*, *Il1rn*, *Vegfa* and *Tgfb1*), chemokines (*Ccl2*, *Ccl6*, *Ccl9*, *Cxcl9*) and chemokine receptors (*Ccr1*, *Ccr2*, *Cxcr4*) in intestinal monocytes/macrophages (Fig 2E [↗](#)). AUC and KEGG analysis demonstrated significant enrichment pathways related to activation of innate immune response and phagocytosis in monocytes/macrophages from BG-treated mice (Fig 2F [↗](#) and 2G [↗](#)). These findings, corroborated by WGCNA analysis, strongly suggest that BG pretreatment enhanced the recruitment and activation of myeloid cells, particularly monocytes/macrophages, in the inflamed colon, thereby improving disease outcomes.

BG trained bone marrow monocytes protected against colitis via the Ccl2-Ccr2 axis

Drawing upon single-cell transcriptome analysis that reveals the proportional expansion and functional reprogramming features of colonic monocyte/macrophage populations, in conjunction with the observation of significantly upregulated peripheral monocytes following BG pretreatment as depicted in Fig S2B [↗](#). A hypothesis was formulated that BG exerts protective effects against colitis via monocytes. To experimentally validate this hypothesis, we evaluated the role of monocytes in *Ccr2* KO mice. Circulating $Ly6C^{hi}$ monocytes were diminished in *Ccr2*^{-/-} mice (Fig S5A [↗](#) and S5B [↗](#)), corroborating the pivotal role of the Ccl2-Ccr2 axis in monocyte egression from the bone marrow into circulation(19, 20). Notably, BG pretreatment failed to protect *Ccr2*^{-/-} mice from DSS-induced colitis, as evidenced by the lack of significant differences in body weight loss, colon length, and epithelial permeability (Fig 3A-E [↗](#)).

In view of the fact that BG induces myelopoiesis through the modulation of hematopoietic stem and progenitors cell compartments in the bone marrow via central trained immunity(21, 22), we assessed whether transplantation of bone marrow cells from BG-pretreated donors could confer resistance to DSS-induced colitis in naïve recipients (Fig 3F [↗](#)). At 6 weeks post-transplantation, circulating myeloid cells were predominately derived from donor mice as indicated by the CD45.1 marker on circulating mononuclear cells (Fig S5C [↗](#)). Following DSS treatment, mice receiving bone marrow cells from BG-pretreated donors exhibited significantly reduced body weight loss (Fig 3G [↗](#)). However, no significant difference in colon length was observed (Fig 3H [↗](#) and Fig S5D [↗](#)). Meanwhile, the percentage of circulating CD11b⁺ myeloid cells, neutrophils and $Ly6C^{hi}$ monocytes were increased (Fig 3I and J [↗](#), Fig S5E [↗](#)).

To further assess the role of monocyte, we adoptively transferred bone marrow $Ly6C^{hi}$ monocytes sorted from control or BG-trained donor mice into *Ccr2*^{-/-} recipient mice, which were then subjected to DSS treatment (Fig S5F [↗](#) and S5G [↗](#)). Mice received BG-trained monocytes exhibited reduced weight loss and colon shortening (Fig 3K [↗](#) and 3L [↗](#)). These findings suggest that BG-trained $Ly6C^{hi}$ *Ccr2*⁺ monocytes play a pivotal role in the alleviation of colitis.

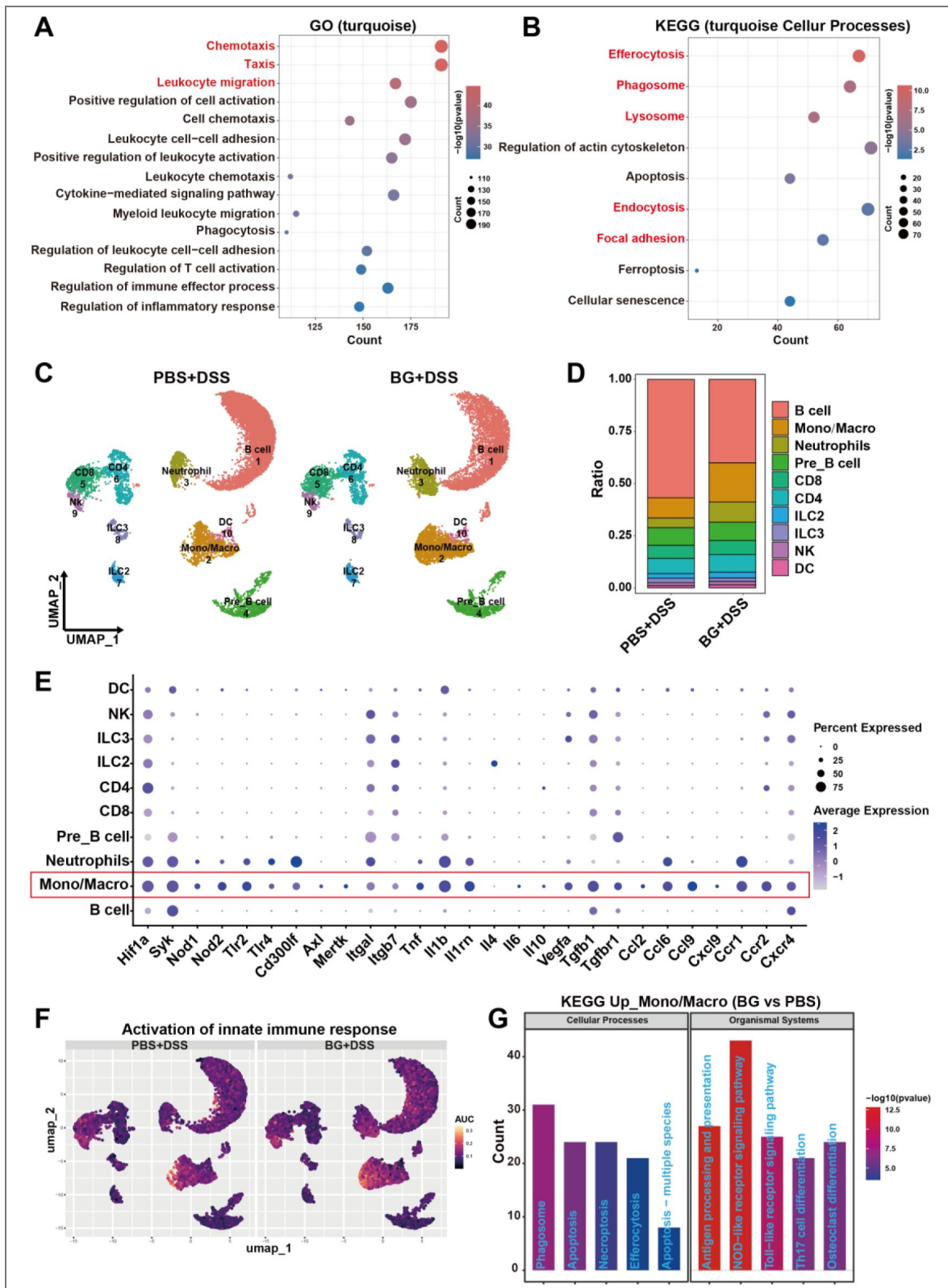


Fig 2. BG ameliorates colitis by enhancing myeloid cell activation.

RNA sequencing of colon tissue at different time points of colitis. (A) GOBP and (B) KEGG pathway analyses of genes in the MEturquoise module. Single-cell RNA sequencing analysis of CD45⁺ cells in the colon on day seven of colitis after one week of BG pretreatment. (C) UMAP plot of LP CD45⁺ cells, (D) Cell ratio distribution from scRNA-seq data, (E) Dot plots showing representative DEGs between LP CD45⁺ cells, (F) AUC scores for selected pathways, (G) KEGG pathway analysis of genes upregulated in the monocyte-macrophage lineage. DEGs, differentially expressed genes. LP, lamina propria. UMAP, uniform manifold approximation, and projection. AUC, area under the curve.

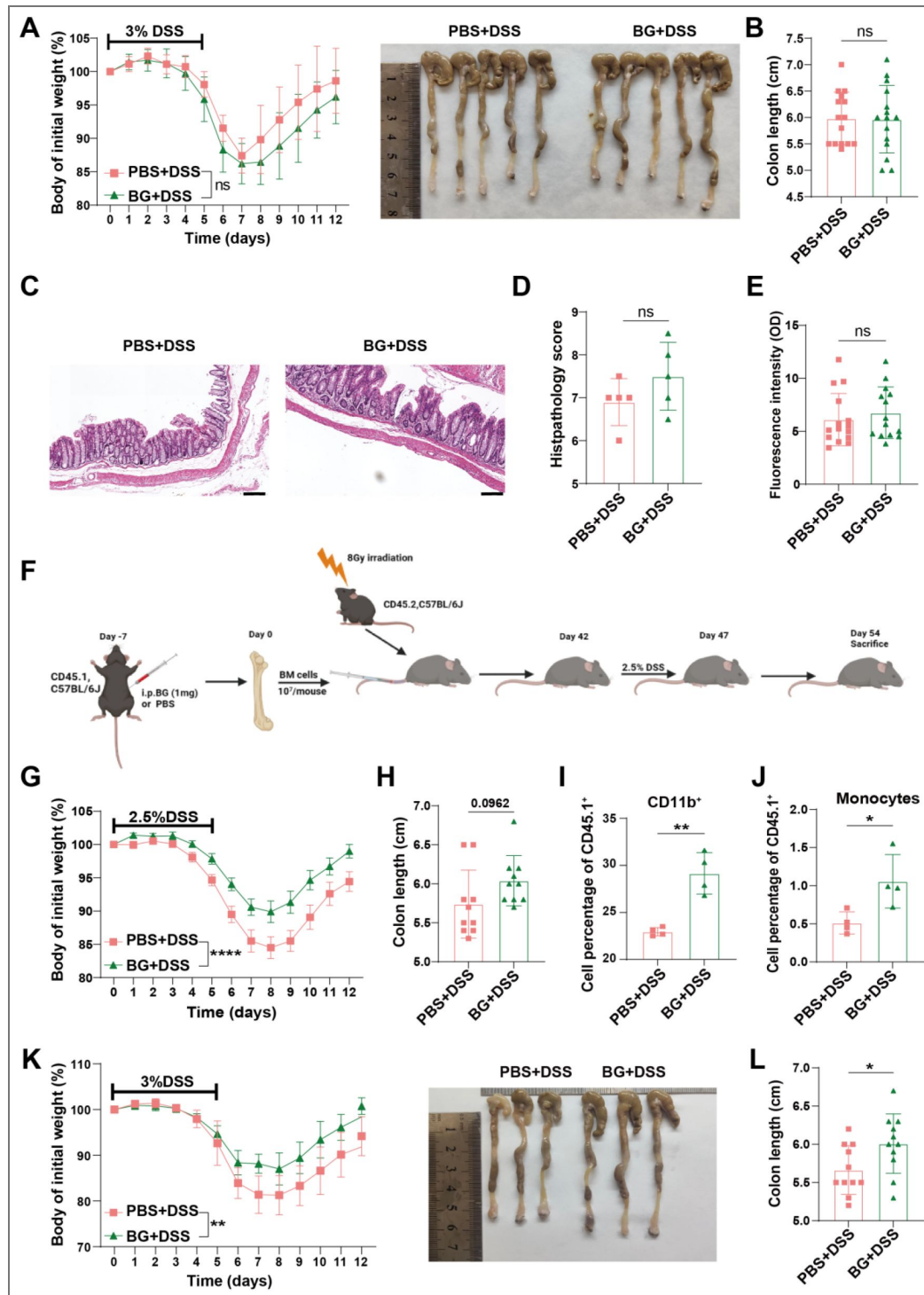


Fig 3. BG trained bone marrow monocytes protected against colitis via the Ccl2-Ccr2 axis.

Ccr2^{-/-} mice pretreated with BG for 1 week followed by induction of colitis with 3% DSS. (A) Changes in body weight ($n = 15$), (B) Colon length changes in *Ccr2*^{-/-} mice from (A), (C and D) H&E staining and histological scoring, H&E. Scale bars: 100 μ m, (E) FITC-dextran assay assessing intestinal barrier function, (F) Schematic representation of bone marrow transplantation from BG-pretreated CD45.1 mice to CD45.2 mice and colitis model, (G) Body weight change curve of colitis mice ($n = 10$), (H) Colon length changes in colitis mice from (G), (I and J) Percentage of CD11b⁺ (I) and Ly6C^{hi} monocytes (J) were analyzed by flow cytometry in peripheral blood, (K and L) Body weight change curve (K) and colon length changes of *Ccr2*^{-/-} mice receiving Ly6C^{hi} monocyte adoptive transfer (L) ($n = 11$). Data are presented as mean $\pm$ SD. Statistical significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.0001$. ns, not significant.

BG-trained monocytes enhance innate immune activation and microbial control

To explore the heterogeneity and functional roles of monocytes and macrophages, we delved deeper into the scRNA-seq data. By examining the expression of key signature genes, including *Ly6c2*, *Ccr2*, *H2-Ab1*, *Runx3*, *Itgax*, *Adgre1*, *Cx3cr1* and *Cd209a*, we identified eight distinct immune cell subsets: three monocyte subsets (Mono1-3) characterized by high *Ly6c2* and *Ccr2* expression, four macrophage subsets (Macro1-4) defined by expression of *H2-Ab1*, *Runx3*, *Itgax*, *Adgre1*, and *Cx3cr1*, and a dendritic cell subset (Cd209a⁺DC) expressing integrin *Itgax* and C-type lectin *CD209a* (Fig 4A and Fig S6A). Among macrophage subsets, Macro1 highly expressed *Vegfa*, suggesting that may be a mucosal repair-promoting macrophage population. Macro2 was enriched with tissue-resident macrophage signature genes (*Runx3*, *Dtx4*, *Cx3cr1*, *Hes1*). In contrast, Macro3 and Macro4 deviated from the main cluster in cell clustering analysis, showing significant heterogeneity. Macro3 was characterized by high *Ighm* expression, while Macro4 showed elevated *Cd4* expression. Both Macro2 and Macro4 exhibited higher *Il10* expression than other monocyte/macrophage subsets, indicating their inflammation-regulatory functions.

The BG-treated group exhibited a significant increase in the proportions of Mono1, Mono2, Macro1, and CD209a⁺DCs, whereas higher proportions of Mono3 and Macro3 were observed in the PBS group (Fig 4B). Given the strong enrichment of Mono1 and the reduction of Mono3 following BG treatment, we sought to further define the transcriptomic characteristics that differentiate Mono1 from Mono3. GO enrichment analysis revealed that Mono1 was significantly enriched in pathways regulating the activation of innate immune response (Fig S6B). KEGG pathways analysis further identified enrichment in NOD-like receptor (NLR) signaling, Toll-like receptor (TLR) signaling, and phagocytosis (Fig 4C). AUC analysis further confirmed the TLR and NLR signaling pathways were significantly upregulated in Mono1 and Mono2 subsets of the BG-pretreated group (Fig S6C and S6D). Additionally, antimicrobial humoral response pathway was also upregulated in these two subclusters (Fig S6E). Cooperative activation of the pattern recognition receptor (PRR) signaling axis and antimicrobial humoral response significantly promotes the synthesis of antimicrobial peptides (23, 24). BG pretreatment significantly upregulated genes encoding pattern recognition receptors (*Nod1*, *Nod2*, *Tlr2*, *Tlr4*) and antimicrobial effector molecules (*S100a8*, *S100a9*, *Lcn2* and *Defb1*) in the colon on day 7 of colitis (Fig 4D). Furthermore, antimicrobial genes *S100a8* and *Defa1* were highly expressed in the BG group as validated by qPCR analysis (Fig S6F). These genes upregulation suggest a potential link between trained immunity and enhanced microbial control. Consistently, gene signatures associated with bacterial defense responses were prominently enriched in the BG-treated group (Fig 4E).

To investigate the epigenetic basis for this sustained transcriptional reprogramming, we reanalyzed publicly available ATAC-seq data from hematopoietic stem cells (HSCs) of BG-trained mice (CRA014389). This analysis revealed that BG training reshaped the epigenomic landscape, with pronounced chromatin opening at promoters and distal intergenic regulatory loci (Fig S7A, B). Functional annotation of these primed regions showed significant enrichment for immune processes, including leukocyte migration and chemotaxis (Fig S7C), as well as pathways like chemokine signaling (Fig S7D). Crucially, promoter-centric analysis highlighted the enrichment of defense response to bacterium (Fig S7E), and locus-specific examination confirmed BG-induced chromatin remodeling upstream of key antimicrobial effectors, including *Gbp2*, *Gbp5*, *S100a8*, and *Nos2* (Fig S7F). This demonstrates that BG establishes a permissive epigenetic architecture in progenitor cells, priming the expression of immune and antibacterial programs.

Consistent with this primed epigenetic state, we observed a significant enrichment of the IFN- γ response pathway (Fig S6G) and a specific upregulation of Guanylate-binding protein (Gbp) genes (*Gbp2*, *Gbp3*, *Gbp5*, *Gbp7*) in the Mono1 and Mono2 subsets (Fig 4F). Previous studies have demonstrated that GBPs as intracellular effectors induced by IFN γ and LPS to promote antibacterial defense (25, 26). GBP-deficient mice exhibit significantly increased susceptibility to *Salmonella typhimurium* infection (27–29). We then performed intestinal *Salmonella typhimurium*

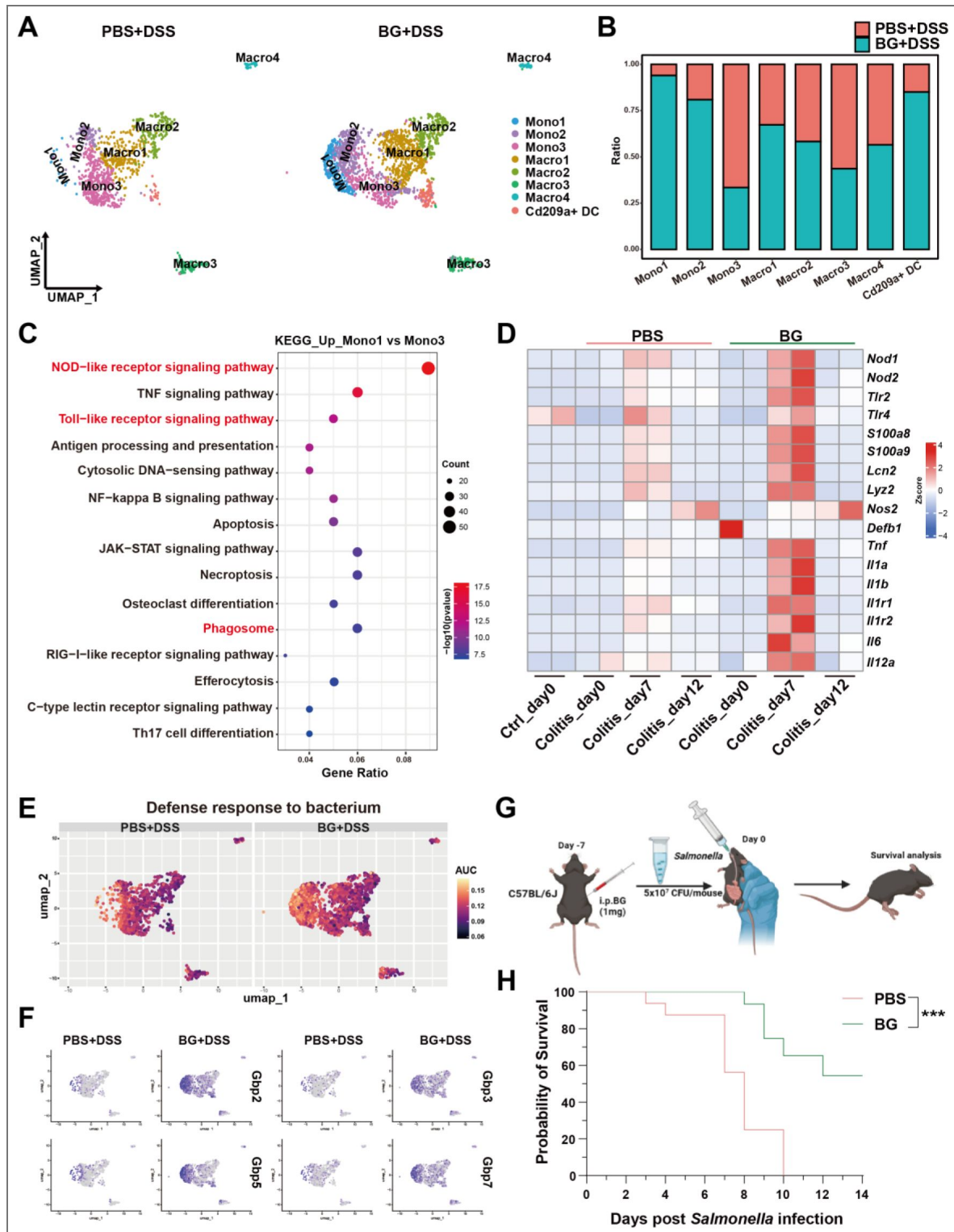


Fig 4. BG-trained monocytes enhance innate immune activation and microbial control.

(A) UMAP and graphical visualization of the monocyte/macrophage lineage, (B) The ratio of monocyte/macrophage subsets, (C) KEGG pathway enrichment analysis of genes upregulated in monocyte 1 and monocyte 3, (D) Gene expression analysis at different time points of colitis, based on colon RNA sequencing, (E) AUC scores of selected pathways, (F) UMAP plots showing differential gene expression patterns, (G) Schematic representation *Salmonella* resistance after one week of BG pretreatment, (H) Survival curve of mice infected with *Salmonella* after one week of BG training. Statistical significance: ***p < 0.001.

infection to assess whether BG pretreatment enhances microbial control in the gut (Fig 4G [↗](#)). As expected, BG pretreatment significantly protected mice from lethal *Salmonella Typhimurium* infection, suggesting that BG-induced trained immunity enhances mucosal defenses against gut microbial infections (Fig 4H [↗](#)).

Taken together, these results reinforced our bulk RNA-seq findings, indicating that BG pretreatment induces a specific monocyte subcluster with enhanced innate activation and phagocytic capacity, facilitating more efficient control of the breaching microbiome associated with DSS-induced leaky gut.

BG-mediated reprogramming of myeloid differentiation trajectories balances inflammation and enhances mucosal repair in colitis

While BG-induced trained immunity is characterized by enhanced pro-inflammatory cytokine production and increased bactericidal capacity([14](#)), excessive pro-inflammatory cytokines release may exacerbate colitis progression. To balance heightened cytokine production without inducing overwhelming immunopathology, we hypothesized that BG pretreatment might involve feedback loops that modulate the inflammation while preserving enhanced phagocytic capacity.

To investigate this, we analyzed previously obtained bulk colonic RNA-seq data, focusing on genes associated with anti-inflammatory and regulatory functions. On day 7 of colitis, BG pretreatment significantly upregulated genes encoding immunomodulatory factors (*Il4*, *Il10*, *Il11* and *Il33*), signaling inhibitors (*Socs1*, *Socs3*), and differentiation regulators (*Csf1*, *Csf1r*, *Stat3* and *Stat6*) (Fig S8A [↗](#)). The upregulation of *Socs1* and *Socs3* was further corroborated by scRNA-seq data, which showed increased expression of these genes in the Mono1 and Mono2 subclusters (Fig S8B [↗](#)). As members of the suppressor of cytokine signaling (SOCS) family, SOCS1 and SOCS3 inhibit excessive JAK-STAT signaling activation via negative feedback regulation([30–32](#)). Moreover, coordinated activation of the monocyte/macrophage differentiation regulatory network (*Csf1r*, *Nr4a2*, *Irf8*, *Klf4*) suggests that BG may induce the differentiation of regulatory-phenotype macrophage subsets by reprogramming myeloid developmental trajectories, thereby modulating inflammatory responses.

To further investigate whether BG pretreatment reprograms monocyte-to-macrophage differentiation, we analyzed the day7 RNA-seq data, which revealed significant enrichment of myeloid leukocyte differentiation pathway (Fig S8C [↗](#)). Additionally, Single-cell trajectory analysis further revealed that Macro1 and Macro2 differentiated from Mono2 and Mono3, respectively (Fig 5A [↗](#)). BG training significantly accelerated the differentiation kinetics of Mono3 into Macro1, which may underlie the reduced frequency of the Mono3 subset in the BG group. Coupled with the gene expression signatures in Fig S6A [↗](#), Macro1 shares similarities with Macro2 in marker gene expression, and the differentiated Macro1 subset could further differentiate into the tissue-resident macrophage Macro2 subset (Fig 5B [↗](#)).

Furthermore, we observed that the expression levels of *Ccr2* and *Ly6c2* decreased from Mono1 to Macro2, while *H2-Ab1* and *Cx3cr1* progressively increased, particularly in macrophages (Fig 5C [↗](#)). Since *Cx3cr1* upregulation is a hallmark of circulating monocytes migrating to the intestinal lamina propria and undergoing macrophage differentiation([33](#)), we used *CX3CR1-GFP* reporter mice to monitor the monocyte-to-macrophage transition in the DSS colitis model. In circulating leukocyte subsets, the total percentage of CD11b⁺ myeloid cells was higher in BG-pretreated mice compared to controls (Fig S9A-C [↗](#)). A similar trend was observed in neutrophils (Fig S9D [↗](#) and S9E [↗](#)). While the initial monocyte percentage was higher in the BG group, no significant difference was observed the two groups on day 7 post-DSS treatment (Fig 5D [↗](#) and S9F [↗](#)), suggesting that circulating monocytes in BG-pretreated mice had extravasated into the intestinal tissue. Meanwhile, the percentage of colonic CD11b⁺ myeloid cells and neutrophils was increased in the BG-pretreated group (Fig S10A-C [↗](#)).

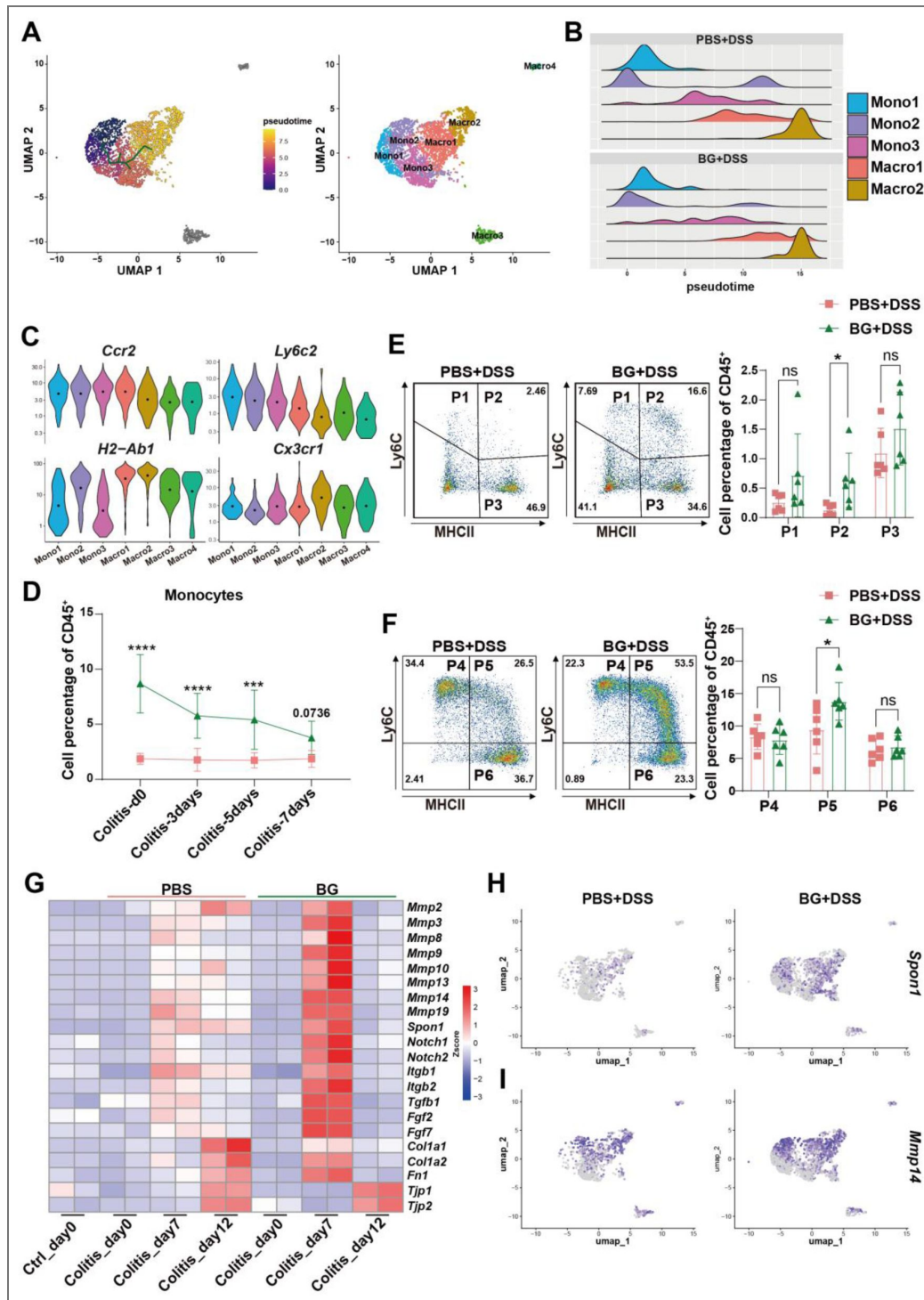


Fig 5. BG-mediated reprogramming of myeloid differentiation trajectories balances inflammation and enhances mucosal repair in colitis.

(A) Monocle 3 trajectory analysis of monocyte/macrophage subsets, (B) Ridgeline plot of monocyte/macrophage subsets, (C) Violin plots of surface marker gene expression in monocyte/macrophage subsets, (D) Percentage of monocytes in peripheral blood at different time points of colitis progression, (E and F) Colonic LPMCs were collected, and the percentages of monocyte/macrophage were analyzed on day 7 of colitis, (G) Expression of mucosal repair-related genes at different time points of colitis, (H) Gene expression analysis of the monocyte/macrophage lineage. Data are presented as mean $\pm$ SD. Statistical significance: * $p < 0.05$, *** $p < 0.001$; **** $p < 0.0001$. ns, not significant.

We further defined monocyte/macrophage populations from P1 to P6 based on the expression of CD11b, CX3CR1-GFP, Ly6C, and MHCII. P1 to P3 were gated from CD11b⁺&CX3CR1-GFP⁺&Ly6C⁺ cells. While P1 (Ly6C⁺&MHCII⁺) and P3 (Ly6C⁺&MHCII⁺) were comparable between group, P2 (Ly6C⁺&MHCII⁺) was increased in BG-pretreated mice (Fig 5E). P4 to P6 are gating from CD11b⁺&CX3CR1-GFP⁺ cells and further separate by the expression of Ly6C and MHCII. P4 (Ly6C⁺&MHCII⁺) as infiltrating monocytes gaining CX3CR1 expression. P5 (Ly6C⁺&MHCII⁺) is regarded as intermediate transitory immature macrophage and P6 (Ly6C⁺&MHCII⁺) is considered as mature macrophage as it completely loss expression of Ly6C and highly express MHCII. BG pretreatment led to an increase in P5 level (Fig 5F). The increase of P2 and P5 in BG group in line with the decrease of monocytes at day 7 in circulation post DSS treatment, suggesting that BG trained monocytes have higher capacity to infiltrate into the damaged colon and undergo macrophage differentiation.

Previous WGCNA analysis suggested that BG pre-treatment upregulated the focal adhesion pathway, which may enhance tissue repair mechanisms (Fig 2B). Consistent with this, we observed increased expression of genes involved in wound healing and tissue repair, such as *Mmp*-related genes, *Spon1*, *Notch1*, *Notch2*, *Tgfb*, *Fgf1*, *Fgf2*, *Col1a1*, *Col1a2*, and *Fn1* in the BG-treated group on day 7, followed by a rapid decline by day 12. In contrast, the PBS group exhibited a much milder induction of these genes, with several genes remaining at comparable levels between day 7 and day 12 (Fig 5G). This led us to hypothesize that the upregulated monocytes and macrophages in the BG-treated group might exhibit enhanced expression of tissue repair-associated genes. As expected, the expression of *Spon1* and *Mmp14*, which promote the mucosal repair was increased in Mono2, Macro1, and Macro2 (Fig 5H and 5I). These findings suggest that BG pretreatment indeed induced monocyte and macrophage subsets with enhanced tissue repair capacity.

Discussion

Trained immunity, a non-antigen-specific form of innate immunological memory, enables rapid host defense mobilization upon pathogen challenge(34, 35). In inflammatory bowel disease, gut microbial dysbiosis manifests as reduced α -diversity, diminished commensal taxa (e.g., *Faecalibacterium prausnitzii*), and pathobiont expansion (e.g., *Enterobacteriaceae*, *adherent-invasive Escherichia coli*), which may compromise intestinal barrier integrity and precipitate microbial translocation(36–38). Additionally, the long-term use of immunosuppressants and biologics in IBD patients elevates infection susceptibility, rendering the microbial-controlling properties of trained immunity a viable therapeutic strategy for IBD(39). However, Excessive or sustained immune activation could trigger chronic inflammatory cascades and provoke autoimmune tissue damage through aberrant immune recognition, as observed in conditions such as rheumatoid arthritis and cardiovascular diseases(9, 40–42). However, the functional correlation between trained immunity and colitis remains unclear, and how to apply trained immunity to colitis treatment represents an urgently unresolved scientific question.

Here, we demonstrated that BG-induced trained immunity confers cross-pathogen protective effects using a lethal *Staphylococcus aureus* infection model, while failing to induce intestinal inflammation under conditions of an intact mucosal barrier. Further investigations revealed that BG effectively attenuated dextran sulfate sodium (DSS)-induced colitis upon secondary challenge. BG pretreatment conferred long-term protection against DSS-induced colitis, with benefits lasting for at least two months after the peripheral myeloid cell populations returned to baseline levels. Bone marrow transplantation experiments confirmed that BG-induced trained immunity persists within hematopoietic progenitors, leading to durable reprogramming of peripheral monocytes. This long-term effect is likely supported by epigenetic modifications in hematopoietic cells, as reported in previous studies(21, 43). These findings highlight the potential of BG-induced trained immunity in establishing long-lasting immune memory, which could be leveraged to achieve sustained therapeutic effects in IBD.

The conventional understanding of IBD pathogenesis has long posited that abnormal immune activation is central to disease development and progression. This paradigm has predominantly focused on the role of adaptive immunity, particularly the dysregulation of Th17 and Treg functions over the past decades(44–46). However, our results suggest that BG-induced protection is independent of adaptive immune responses. Even in *Rag1*^{−/−} mice, which lack adaptive immunity, BG pretreatment significantly improved DSS-induced colitis outcomes. *Rag1*^{−/−} mice retain ILCs (including ILC3s), which promote mucosal repair via IL-22 production(47). While BG-induced ILC activation remains untested, our monocyte adoptive transfer experiments show monocytes alone alleviate colitis—suggesting a dominant, but not exclusive, role for monocytes/macrophages. ILC contributions cannot be ruled out and warrant further investigation.

In DSS-induced colitis, epithelial barrier disruption facilitates microbiota translocation across the mucosal epithelium, triggering inflammatory responses mediated by innate cells(38, 48, 49). We identified monocytes as key mediators of BG-induced protection against DSS-induced colitis, as demonstrated by experiments using *Ccr2*^{−/−} mouse models and monocyte transfer studies. BG pretreatment enriched Ly6C^{hi}CCR2⁺ monocytes, which exhibited enhanced antimicrobial capabilities, as revealed by scRNA-seq. Ly6C^{hi}CCR2⁺ monocytes displayed higher levels of antimicrobial receptors like NOD2, TLR4 and GBP-related gene. Guanylate-binding protein (GBP) enhances *Salmonella* clearance by promoting inflammasome activation and intracellular bacterial elimination, particularly through disrupting *Salmonella*-containing vacuoles and inducing pyroptosis(27, 50). Our finding revealed that BG pretreatment increased resistance to *Salmonella typhimurium* infection. As inflammation and infection is partially controlled, monocytes can differentiate into macrophage subsets with anti-inflammatory and tissue repair functions, secreting cytokines such as IL-10, TGF-β and VEGF to promote inflammation resolution and tissue repair(51, 52).

Previous studies have underscored the essential role of circulating Ly6C^{hi} monocytes in replenishing intestinal macrophages(53, 54). During colitis and infection, these monocytes are rapidly recruited to injured colonic tissues, where they differentiate into Cx3cr1⁺ macrophages. We also observed that these monocytes exhibited enhanced migratory capabilities, enabling them to infiltrate inflamed tissues and differentiate into CX3CR1⁺ macrophages. These macrophages contribute to mucosal repair by clearing pathogens such as *Salmonella* and promoting epithelial regeneration(55). Single-cell trajectory analysis further showed that BG training promoted the differentiation of Macro1 into Macro2, which is consistent with the dynamic process of monocyte-to-mature macrophage conversion observed in *CX3CR1-GFP* reporter mice. Maintaining a balance between inflammatory monocytes, differentiated macrophages, and tissue-resident macrophages is critical for resolving inflammation, promoting tissue repair(33, 56). These findings suggest that BG not only augments monocyte antimicrobial functions but also facilitates their differentiation into macrophages, enhancing their ability to recognize, phagocytose, and eliminate bacteria while promoting tissue repair. This dual function likely strengthens the intestinal barrier, mitigates inflammation, and supports gut homeostasis.

Notably, our findings align with emerging evidence that trained immunity (TI) is not a uniform biological process but encompasses distinct functional programs shaped by the initiating stimulus and contextual cues—including immunoregulatory and tissue-reparative responses. Yang Li et al have identified discrete monocyte/macrophage subpopulations following TI induction, such as cytokine-chemokine dual-high (MCI) and chemokine-high (MC) subsets, which exhibit distinct pathway enrichments and disease associations(57). Consistent with this, β-glucan (BG), the TI inducer in our study, has been shown to drive a specific subset of macrophages with M2-like phenotypic traits (e.g., high CD206 expression, increased IL-10 secretion) while retaining potent antimicrobial capabilities (e.g., bacterial killing, pro-inflammatory cytokine release upon LPS stimulation)—reflecting an immunoregulatory TI program that balances pathogen clearance and inflammation resolution(58). Furthermore, BG-induced TI has been reported to inhibit NLRP3 inflammasome activation via blocking K⁺ efflux and mitochondrial ROS generation, highlighting its context-dependent anti-inflammatory potential in autoinflammatory diseases(59)—a mechanism that may contribute to its protective effects in colitis by limiting excessive intestinal

inflammation. Our findings of BG-induced protection against DSS colitis, characterized by enhanced microbial clearance and accelerated mucosal repair without exacerbating inflammation, represent a paradigmatic example of such context-dependent, reparative TI programs. This aligns with the broader understanding that TI can be tailored to elicit tissue-protective effects when triggered by appropriate stimuli, as demonstrated by BG's ability to prime myeloid cells for both antimicrobial defense and tissue repair(60). Together, these observations place our work within the growing framework of TI functional heterogeneity, emphasizing that the biological outcome of TI is dictated by the interplay between the inducing stimulus, target cell type, and local microenvironment—an insight that further clarifies the mechanism by which BG-induced TI confers colitis protection.

In the clinical exploration of IBD treatment, novel therapies based on cellular regeneration (intestinal organoid transplantation) and fecal microbiota transplantation (FMT) are emerging as critical avenues to overcome the limitations of conventional pharmacotherapies(61–63). Given the efficacy of BG-induced trained immunity in alleviating DSS-induced colitis, this approach holds significant promise as a novel therapeutic strategy for IBD treatment. We transferred BG-preconditioned monocytes which significantly alleviated clinical symptoms, including weight loss and colon shortening, demonstrating the potential of cell-based therapies (Fig S11A–C).

In conclusion, BG confers long-term colitis amelioration through trained immunity by reprogramming bone marrow-derived monocytes. Mechanistically, BG enhances monocyte chemotaxis and antimicrobial functionality activation, enabling efficient microbial containment. Crucially, this trained state establishes self-regulating circuits, engaging in negative feedback regulation of excessive immune responses, while accelerated CX3CR1⁺ macrophage differentiation facilitates mucosal repair. These findings establish a foundation for leveraging trained immunity as a therapeutic strategy for IBD, highlighting the potential of BG in modulating intestinal immunity and restoring gut homeostasis.

Limitations

Although our reanalysis of publicly available ATAC-seq datasets (data available from CRA014389) targeting monocytes confirmed BG-induced chromatin accessibility enrichment at antibacterial defense related genomic regions, direct mechanistic evidence for epigenetic and metabolic profiling in our colitis model remains lacking, and we plan to pursue as future work. Moreover, the functional role of CX3CR1⁺ macrophages remained correlative. Despite transcriptional evidence of repair-related gene expression and phenotypic correlation with colitis resolution, we lack direct causal data due to the unavailability of specific tools for this subset. The fate of adoptively transferred monocytes was not directly traced. Current evidence remains correlative and requires validation through future lineage-tracing studies.

Materials and Methods

Animals

C57BL/6J wild-type male mice (6–8 weeks) were purchased from Xiamen University Laboratory Animal Center. *Rag1*^{−/−} and *CX3CR1-GFP* mice were generously provided by Dr. Kairui Mao and Dr. Xiaofen Chen from Xiamen University, respectively. *Ccr2*^{−/−} mice were kindly provided by Dr. Shih-Chin Cheng from Xiamen University. Mice were maintained under specific pathogen-free conditions. All animal protocols were reviewed and approved by the Institutional Animal Care and Use Committee of Xiamen University.

In vivo beta-glucan (BG) training and DSS colitis model

Mice were i.p. injected with one dose of BG (1 mg) on day −7/−49. Then, Mice received 3% indicated DSS (36,000 to 50,000 molecular weight; MP Biomedicals) in their drinking water for 5 days, followed by 2 days or 7 days of distilled water without DSS. Control animals received distilled water for the entire period. Mice were monitored every day from day 0 for body weight until day 7 or 12.

Histology

Colons were collected from sacrificed mice at the end of each treatment schedule, as described above, colon was fixed with 4% paraformaldehyde and embedded in Paraffin. Tissue sections (5 μ m) were prepared, deparaffinized, and stained with hematoxylin and eosin. With minor modifications, by assessing mucosa thickening, inflammatory cells, and submucosa cell infiltration. Each criterion was scored as 0-4, and the sum of each score was defined as the histological score. Histological scores were assigned by experimenters “blinded” to sample identity.

Coloscopy

A coloscopy system (ENDOCAM® Logic HD, RICHARD WOLF) was used to monitor the severity of DSS-induced colitis. After an 8-hour fast, we administered isoflurane (RWD, Co. Ltd.) and applied glycerin to the anus for smooth insertion.

In vivo intestine permeability assay

Age and sex-matched mice were orally administered with 0.6 mg/g body weight of an 100 mg/ml solution of FITC-dextran (FD4, Sigma). 5 hours later, retro-orbital blood was collected from each mouse. Serum was prepared by allowing the blood to clot by leaving it undisturbed overnight at 4°C and then subsequently centrifuged at 5000 rpm for 10 minutes. Dilutions of FITC-dextran in PBS and separately in pooled mouse serum were used as a standard curve. Absorbance of 50 μ l serum was measured at microplate reader with excitation and emission filters set at 490 and 530 nm, respectively. Experiments were performed at least two independent times each in triplicate.

RNA extraction and real-time PCR

Total RNA was extracted from colon tissue previously washed from luminal content. Samples were homogenized in TRIZOL. Homogenized tissues were then added with chloroform in order to separate RNA from genomic DNA and proteins. RNA isolation and purification required isopropanol (RNA precipitation) and 70% ethanol (RNA wash). Total RNA was resuspended in Nuclease-free water. Subsequently, 1 μ g of RNA was reverse-transcribed using the Hifair® II 1st Strand cDNA Synthesis SuperMix for qPCR (gDNA digester plus) following manufacturer’s instructions. cDNA was analysed through quantitative Real-Time PCR.

Isolation of cells from lamina propria

To isolate lamina propria mononuclear cells (lamina propria MCs), extraintestinal fat tissue was carefully removed and colons were then flushed of their luminal content with physiologic solution, opened longitudinally and cut into 1 cm pieces. Epithelial cells and mucus were removed by incubation with 10 mL of D-Hank’s balanced salt solution (containing EDTA and DTT, free of Ca^{2+} and Mg^{2+}) for 20 minutes at 37°C at 200 rpm in a constant temperature shaker. Colon pieces were then digested in Hank’s (with Ca^{2+} and Mg^{2+}) containing 0.5 mg/ml Collagenase IV and 15 μ g/ml DNase I, for 40 min at 37°C shaking at 200 rpm. The remaining cells were centrifuged and resuspended in FACS buffer (1% FBS, 2 mM EDTA in PBS).

Bone Marrow Transplantation

Wild type CD45.2 mice were subjected to 8 Gy irradiation (RAD SOURCE, RS2000). These irradiated recipients were then divided into three groups: a sham-transplant control group that received an intravenous injection of PBS alone, a control BMT group that received bone marrow cells from naïve CD45.1 donors, and a BG-trained BMT group that received bone marrow cells from CD45.1 donors pretreated with β -glucan. Bone marrow cells were harvested from the femurs of CD45.1 wild-type or BG-trained donor mice. A total of 1×10^7 bone marrow cells from the respective donors were transferred to the irradiated CD45.2 recipients via tail-vein injection. Successful

engraftment was confirmed by flow cytometric analysis of donor-derived (CD45.1⁺) cells in peripheral blood (Fig. S5C [↗](#)). Six weeks post-transplantation, mice were administered 2.5% DSS in drinking water to induce colitis and were monitored daily until day 12.

Adoptive transfer of bone marrow monocytes

BG trained mice were sacrificed, and the femurs were used to harvest bone marrow cells. Bone marrow cells were stained with the following panel: Biotin antibody (B220, CD4, CD8, NK1.1, Ly6G), Live/dead (Cat. L34976, Invitrogen), CD45 (Cat. 45-0451-82, Clone. 30-F11, ebioscience), CD11b (Cat. 48-0112-82, Clone. M1/70, ebioscience), Ly6C (Cat. 128016, Clone. HK1.4, Biolegend). Biotin-labeled lineage antibodies were used to stain bone marrow cells at 4°C for 20 min and then incubated with fluorescence conjugated streptomycin together with other surface protein antibodies for another 20 min. Ly6C^{hi} monocytes were sorted to > 97% purity using a BD Aria III cell sorter.

RNA-seq

RNA was extracted from colon LPLs of individual mice. RNA integrity was assessed using the RNA Nano 6000 Assay Kit of the Bioanalyzer 2100 system (Agilent Technologies, CA, USA). The clustering of the index-coded samples was performed on a cBot Cluster Generation System using TruSeq PE Cluster Kit v3-cBot-HS (Illumina) according to the manufacturer's instructions. The library was sequenced on an Illumina Novaseq platform and 150 bp paired-end reads were generated.

scRNA-seq

Isolated lamina propria mononuclear cells, sample from three mice were pooled together and analyzed per condition. For scRNA-seq of Colon CD45⁺ cells were sorted by BD Arial III. These cells were barcoded with 10× Cellplex oligos before being encapsulated using the 10X Chromium 3' Reagent Kits v3 according to manufacturer's instructions.

RNA-seq data processing and analysis

Raw data of fastq format were quality checked through fastqc software. Gencode and gene model annotation files were downloaded from genome (mm10) website directly. Index of the gencode was built using Hisat2 v2.1.0 and paired-end clean reads were aligned to the gencode using Hisat2 v2.1.0. For differential expression analysis, raw count data (generated via featureCounts) were used as input for DESeq2, with differential genes filtered by |fold change| ≥ 2 and adjusted P < 0.05. FPKM (Fragment per kilobase per million) values were only used for visualization purposes (e.g., heatmaps, clustering analyses), not for between-sample statistical comparisons. Weighted correlation network analysis (WGCNA) was performed using an R package from horvath. All differentially expressed genes were plotted with the “pheatmap” and “ggplot2” R packages. For KEGG enrichment analysis, p-value < 0.05 was used as a threshold to determine significant enrichment for gene sets with the “clusterProfiler” R package.

scRNA-seq data processing and analysis

Raw sequencing data was processed and aligned mm10 mouse reference genome with CellRanger (10× Genomics) v 7.2.0. Resulting filtrated matrices (count matrices) of molecular counts were used as input for further processing with Seurat package V5.0.1 running under R studio. First, quality control was performed to create Seurat object with min features > 200 and removal of cells having < 200 or > 8000 expressed genes or > 5% mitochondrial counts. The total number of recovered cells was mentioned before. Variable features using FindVariableFeatures (using RNA and vst as an assay and selection method as parameters) and normalization using normalization.method = “LogNormalize”, scale.factor = 10000 were performed. This integrated data is scaled using ScaleData function using all genes and “RNA” as assay method.

Doublet cells were filtered by DoubletFinder v3. Principal Component Analysis was performed on variable features using RunPCA and first 30 PCs were chosen. The nearest neighbors using the FindNeighbors(dims=1:12) and FindClusters(resolution=0.8) function. For visualization, the non-

linear dimensional reduction such as UMAP analysis was using RunUMAP function from Seurat. To calculate pseudotime based on the Mono/Marco subset data using the monocle3.

ATAC-seq data analysis

The ATAC-seq data were obtained from the CNCB (China National Center for Bioinformation) under accession number CRA014389. Quality control was performed using FastQC. The paired-end reads were then aligned to the mouse reference genome (mm10) using Bowtie2 with the parameters: -k 1 -D 20 -R 3 -N 1 -L 20 -i S,1,0.50 -X 2000. The aligned reads were converted to sorted BAM files using Samtools. Peak calling was conducted with MACS2 using a q-value threshold of 0.05, a model fold range of 5-50, and the “keep-dup all” option to retain all tags. Peak quantification was performed using Bedtools and normalized to reads per kilobase per million mapped reads (RPKM). Visualization files in BigWig format were generated using Deeptools. The identified peaks were annotated using the ChIPseeker package, and pathway analysis was carried out with the ClusterProfiler package. Gene Set Enrichment Analysis (GSEA) was performed using the MSigDB database (version v2022.1.Mm).

Statistical Analysis

Statistical analyses were conducted using GraphPad Prism version 9.0 and R statistical software (version 4.2.2, R Foundation for Statistical Computing). Data were analyzed using a two-tailed Student's *t*-test. Multiple-group comparisons were performed using one-way or two-way analysis of variance (ANOVA) followed by Tukey's multiple-comparisons test. Survival data were assessed using Kaplan-Meier survival plots, followed by the log-rank test. Significant differences were indicated by an asterisk ($p < 0.05$).

Supplementary figures

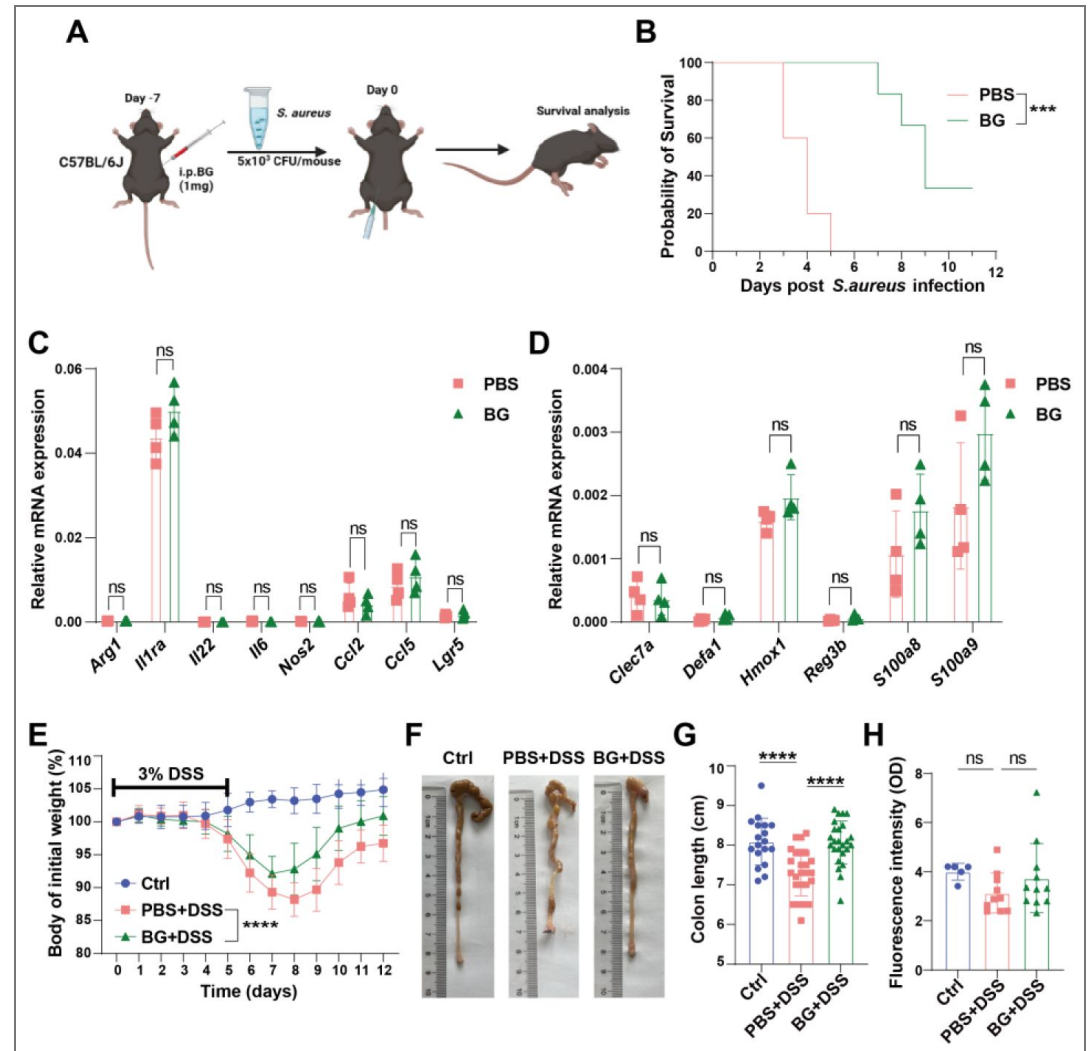


Fig S1. BG pretreatment ameliorates DSS-induced colitis. (A) Schematic representation of BG-induced trained immunity and *Staphylococcus aureus* infection model, (B) Survival curve of mice infected with *Staphylococcus aureus*, (C and D) The gene expression of inflammatory mediators and antibacterials were analyzed by qRT-PCR, (E) Body weight change curve of mice pretreated with BG for one week, followed by colitis induction with 3% DSS ($n = 18-25$), (F and G) Changes of colon length from (E), (H) FITC-dextran assay assessing intestinal barrier function ($n = 5-11$). Data are presented as mean $\pm$ SD. Statistical significance: * $p < 0.05$, *** $p < 0.001$; **** $p < 0.0001$. ns, not significant.

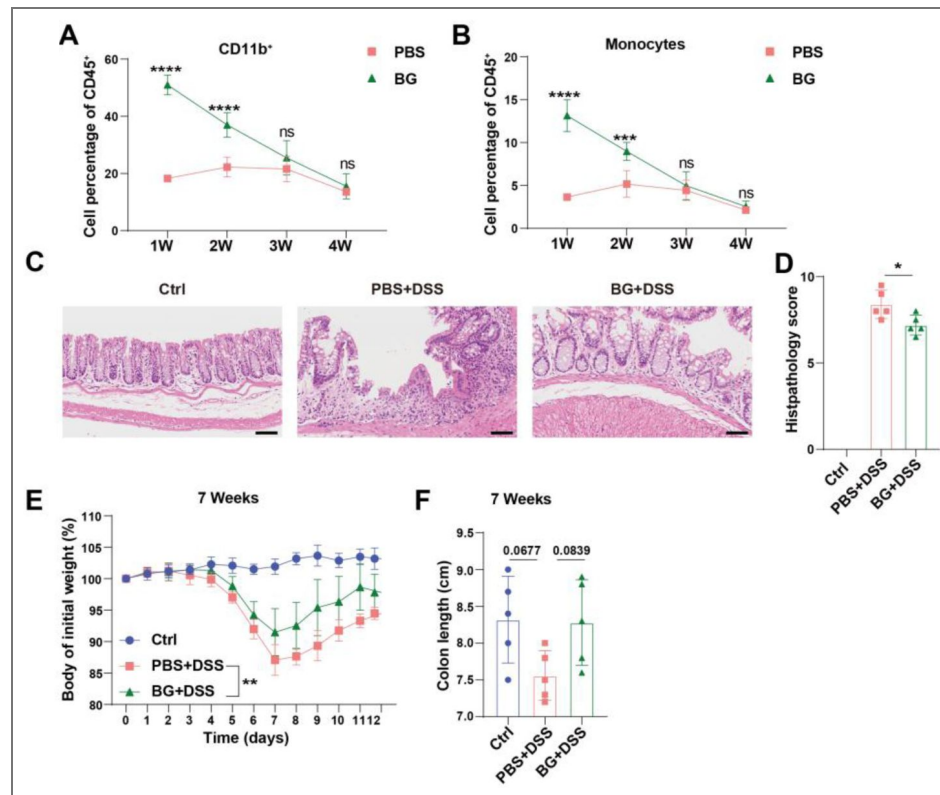


Fig S2. BG pretreatment 4 or 7 weeks ameliorates DSS-induced colitis.

(**A and B**) Percentage of myeloid cells and monocytes in the peripheral blood of mice at different time points after one week of BG pretreatment, (**C and D**) H&E staining, and histological scoring of colitis mice after four weeks of BG training. Scale bars: 100 μm, (**E**) Body weight change curve of mice pretreated with BG for seven weeks, followed by colitis induction with 3% DSS, (**F**) Changes of colon length from (**E**) ($n = 5$). Data are presented as mean $\pm$ SD. Statistical significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. ns, not significant.

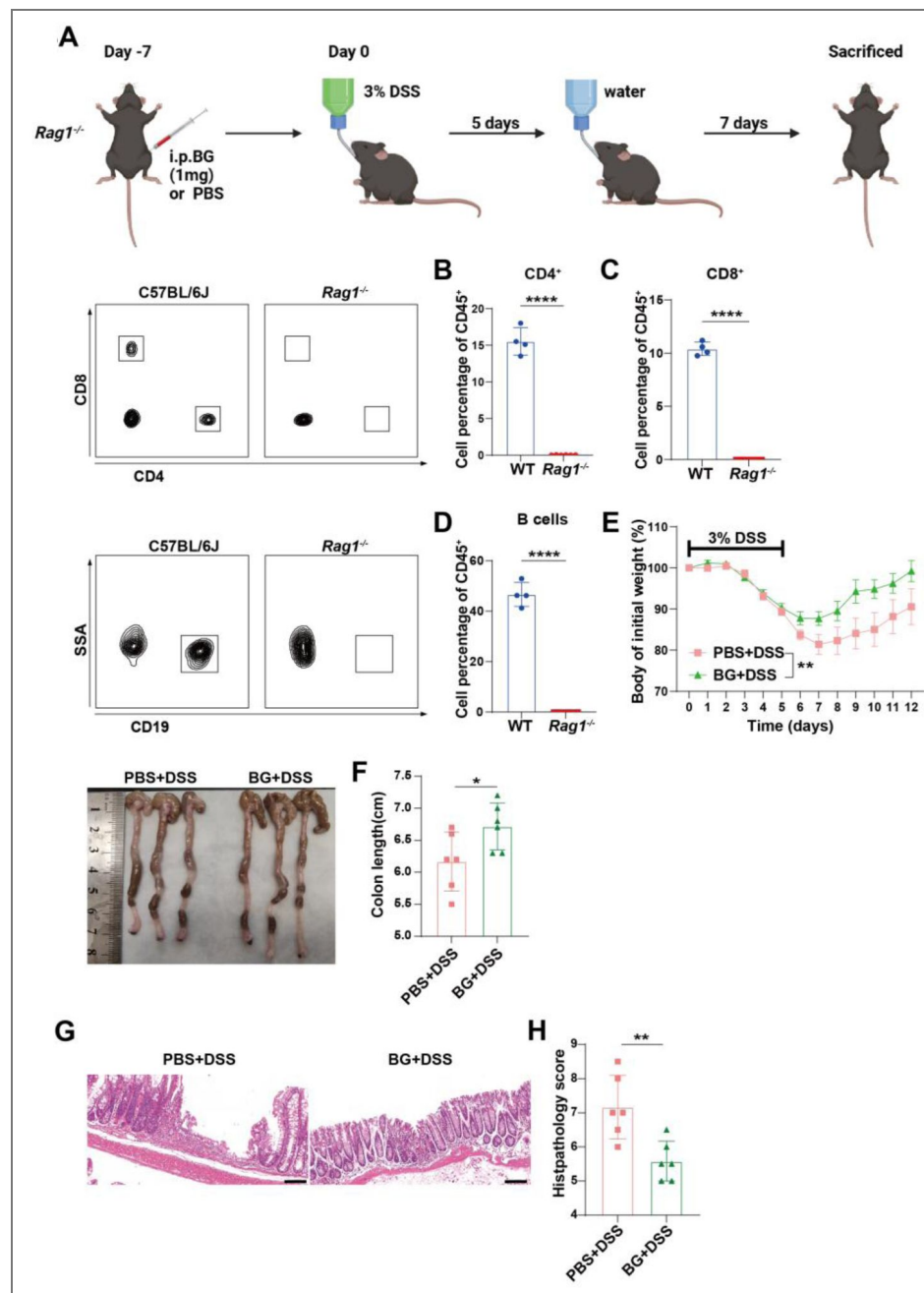


Fig S3. BG pretreatment ameliorates colitis is independent of adaptive immunity.

(A) Schematic representation of BG-induced trained immunity and DSS colitis model in *Rag1*^{-/-} mice, (B-D) Flow cytometry analysis of peripheral blood from *Rag1*^{-/-} mice, (E) Body weight change curve of *Rag1*^{-/-} mice pretreated with BG for one week, followed by colitis induction with 3% DSS, (F) Colon length changes in *Rag1*^{-/-} mice from (E), (G and H) H&E staining and histological scoring. Scale bars: 100 μ m. Data are presented as mean $\pm$ SD. Statistical significance: **p* < 0.05, ***p* < 0.01, *****p* < 0.0001.

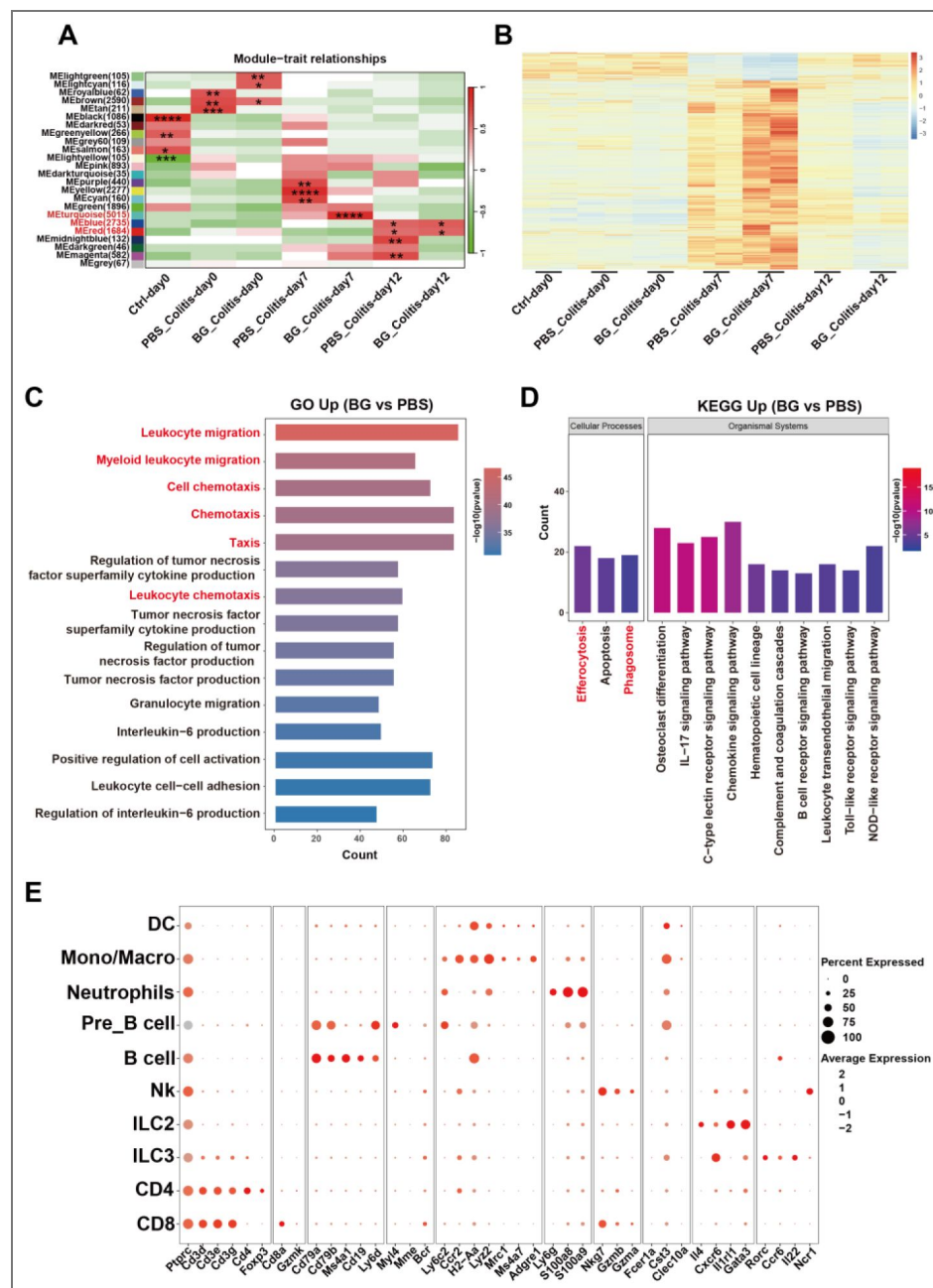


Fig S4. BG ameliorates colitis by enhancing myeloid cell activation.

(A) RNA sequencing of colon tissue at different time points during colitis, with WGCNA identifying major gene modules, (B) DEGs in the Meturquoise module, (C and D) GO and KEGG pathway analysis of colon tissue on day seven of colitis after one week of BG pretreatment, (E) Expression of marker genes in LP CD45⁺ cells on day seven of colitis after one week of BG pretreatment.

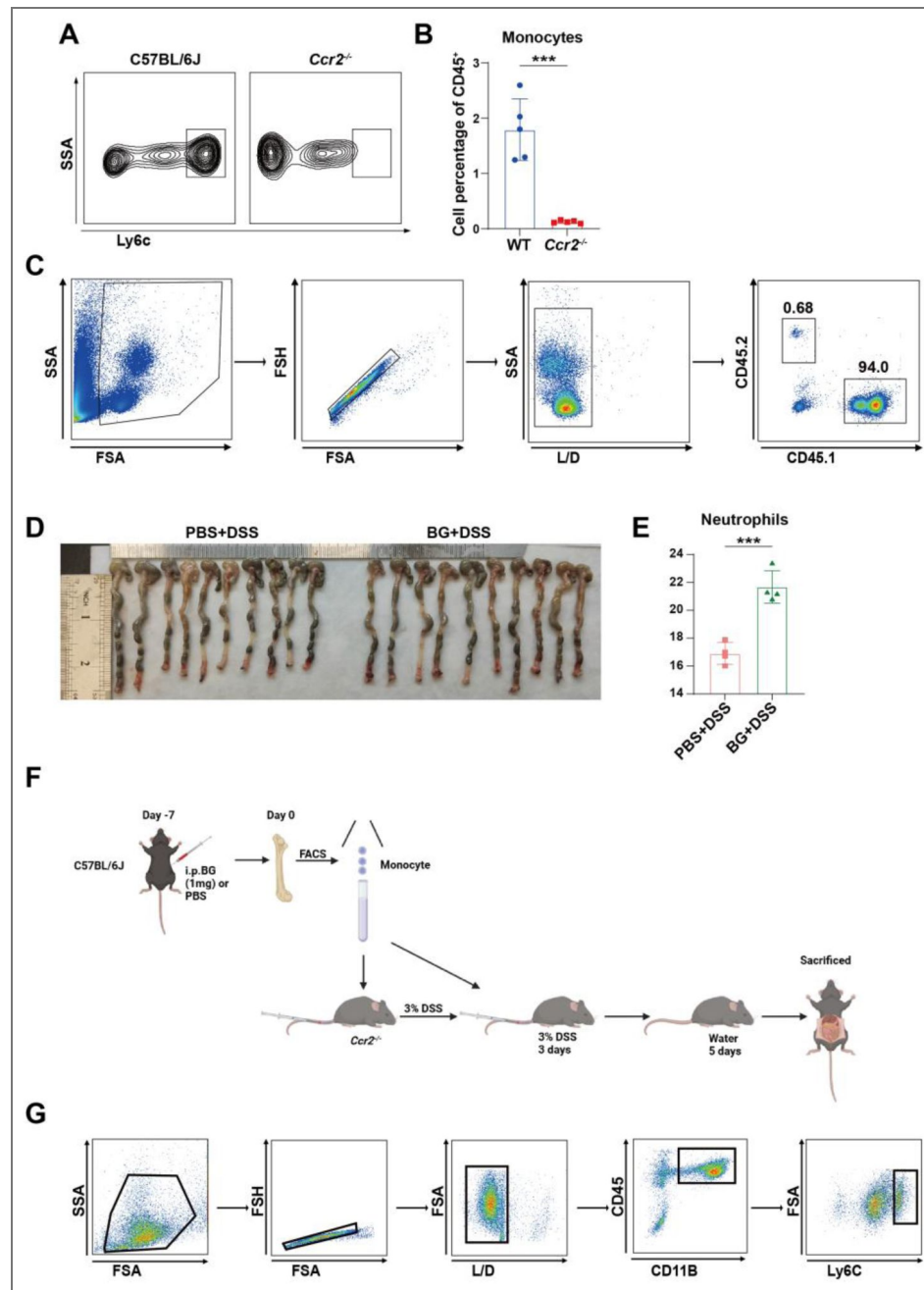


Fig S5. BG trained bone marrow monocytes protected against colitis via the Ccl2-Ccr2 axis.

(A and B) Flow cytometry analysis of peripheral blood from *Ccr2*^{-/-} mice, (C) Flow cytometry analysis of bone marrow reconstitution in CD45.2 recipient mice, (D) Colon length changes in bone marrow transplantation experiments from BG-pretreated CD45.1 mice to CD45.2 mice, (E) Percentage of Ly6G⁺ neutrophils in peripheral blood of CD45.2 recipient mice, (F) Bone marrow monocytes transplant model, (G) Flow sorting scheme of bone marrow monocytes. Data are presented as mean ± SD. Statistical significance: ***p < 0.001.

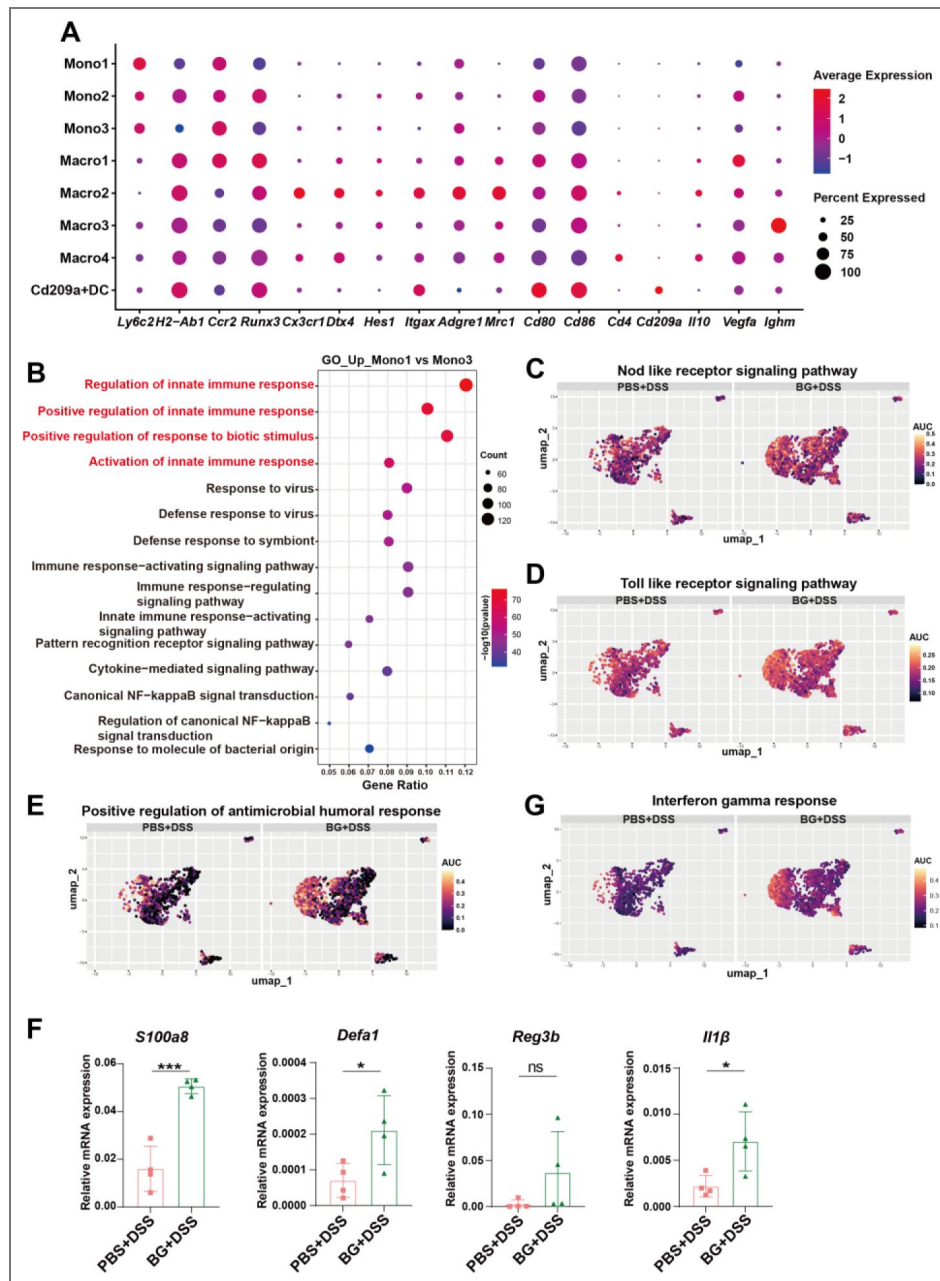


Fig S6. BG pretreatment enhances innate immunity and phagocytic capacity.

(A) Major marker genes of eight cell clusters within the monocyte/macrophage lineage, (B) GO analysis of genes upregulated in monocytes 1 and monocyte 3, (C-E) AUC scores of selected pathways, (F) Antibacterial gene expression was analyzed by qRT-PCR. (G) AUC scores of selected pathways. Data are presented as mean $\pm$ SD. Statistical significance: * $p < 0.05$, *** $p < 0.001$, ns, not significant.

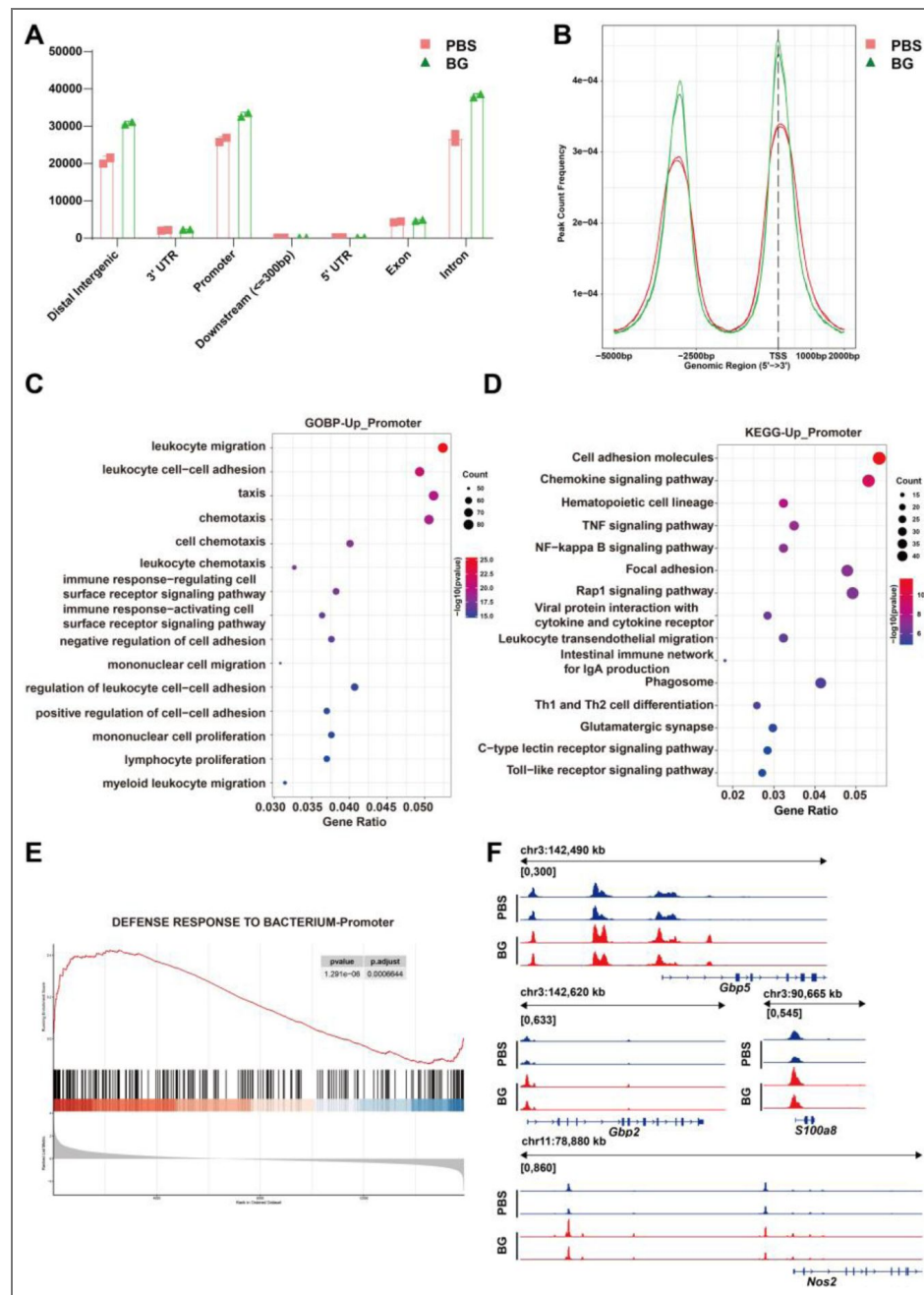


Figure S7. BG induced immune-related epigenomic remodeling in monocytes.

This figure delineates β -glucan (BG)-mediated epigenomic reprogramming in PBS (control) vs. BG-treated monocytes. (A) Bar plot of chromatin accessibility signal enrichment across genomic regions. Bars denote signal counts, showing BG-altered region-specific signal accumulation. (B) Density plot of chromatin peaks relative to promoter TSS (± 2000 bp). BG shifts peaks from TSS core to promoter flanks, indicating promoter-proximal epigenomic reorganization. (C) GO BP bubble plot for genes with BG-upregulated promoter accessibility. Enriched terms focus on immune processes. (D) KEGG pathway bubble plot for enriched immune/inflammatory pathways support BG's role in trained immunity. (E) Enrichment profile and heatmap for the "defense response to bacterium" promoter module, highlighting BG-induced epigenetic priming of antibacterial programs. (F) Locus-specific chromatin accessibility tracks for selected immune genes, validating BG-induced epigenetic modifications in upstream regions relative to PBS controls.

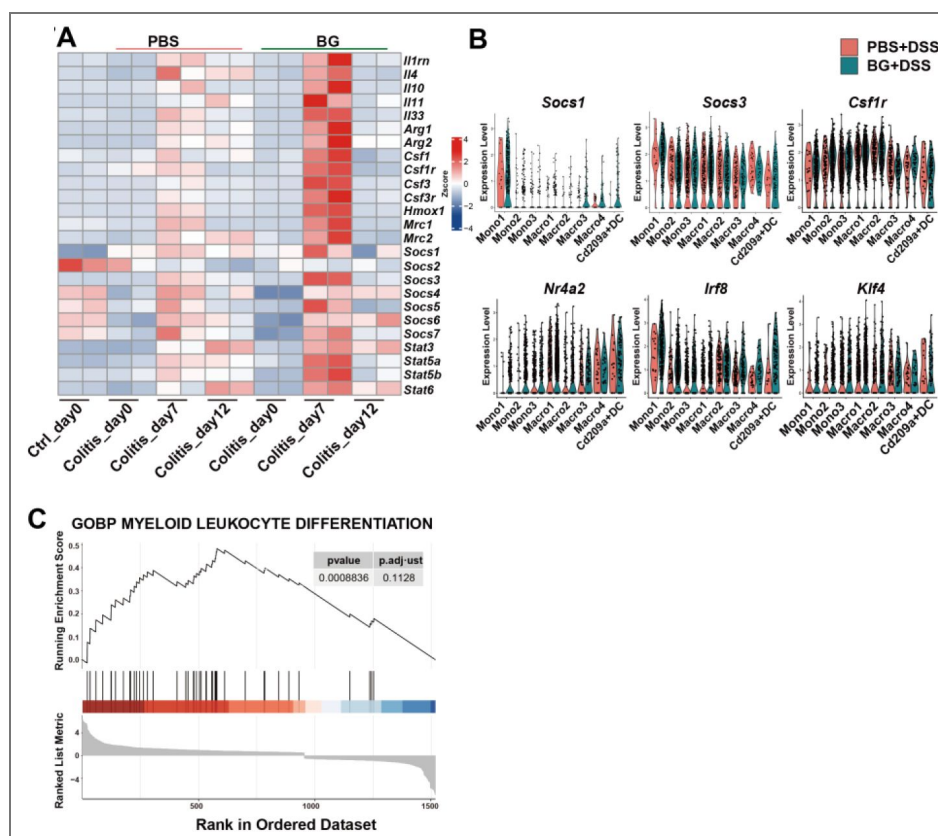


Fig S8. BG-induced trained immunity promotes monocyte differentiation.

(A) Gene expression analysis at different time points of colitis, based on colon RNA sequencing, (B) Violin plots of gene expression in eight cell clusters, (C) GSEA analysis showing gene enrichment pattern on day seven of colitis.

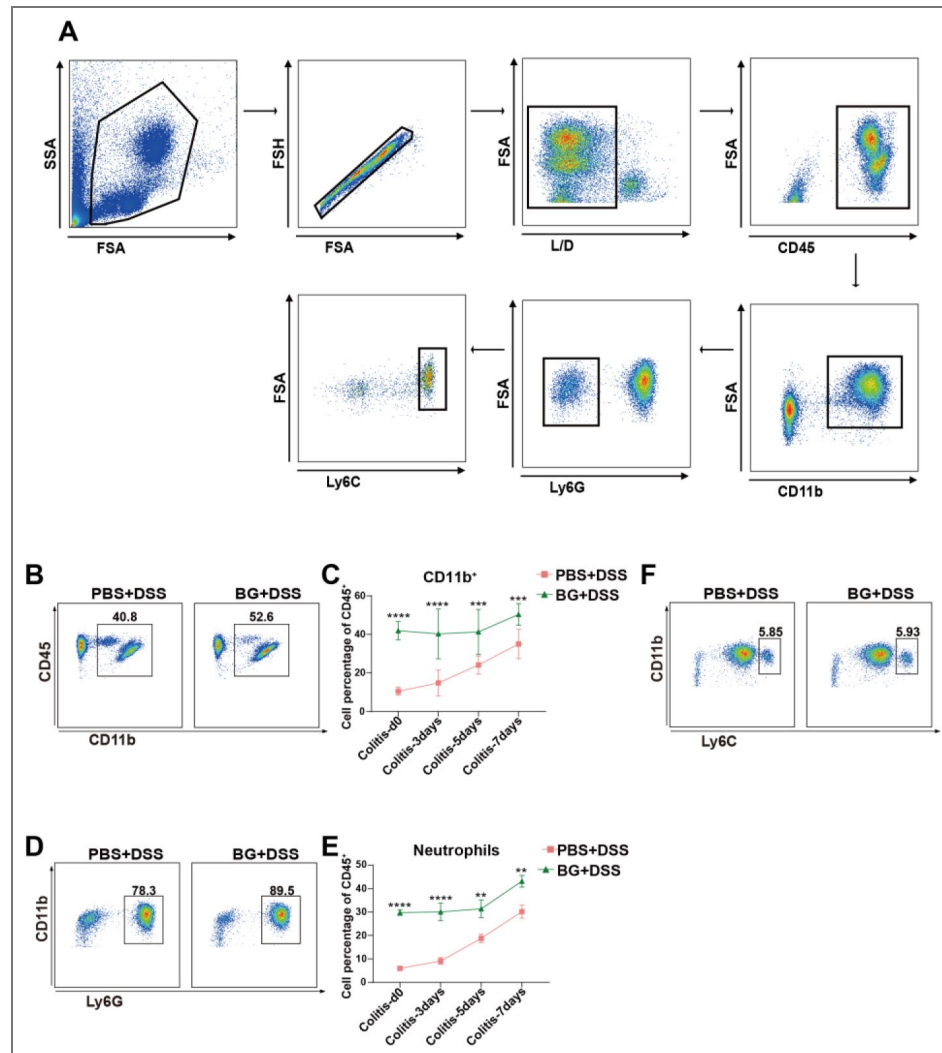


Fig S9. BG training downregulates the proportion of peripheral monocytes during colitis development.

(A) Peripheral blood flow cytometry gating strategy, (B) Representative flow cytometry plots of CD11b⁺ myeloid cells in peripheral blood on day seven of colitis (C) The percentage of CD11b⁺ myeloid cells at different time points of colitis, (D) Representative flow cytometry plots of Ly6G⁺ neutrophils in peripheral blood on day seven of colitis, (E) Percentage of Ly6G⁺ neutrophils at different time points of colitis, (F) Representative flow cytometry plots of Ly6C^{hi} monocytes in peripheral blood on day seven of colitis. Data are presented as mean $\pm$ SD. Statistical significance: ** p < 0.01; *** p < 0.001; **** p < 0.0001.

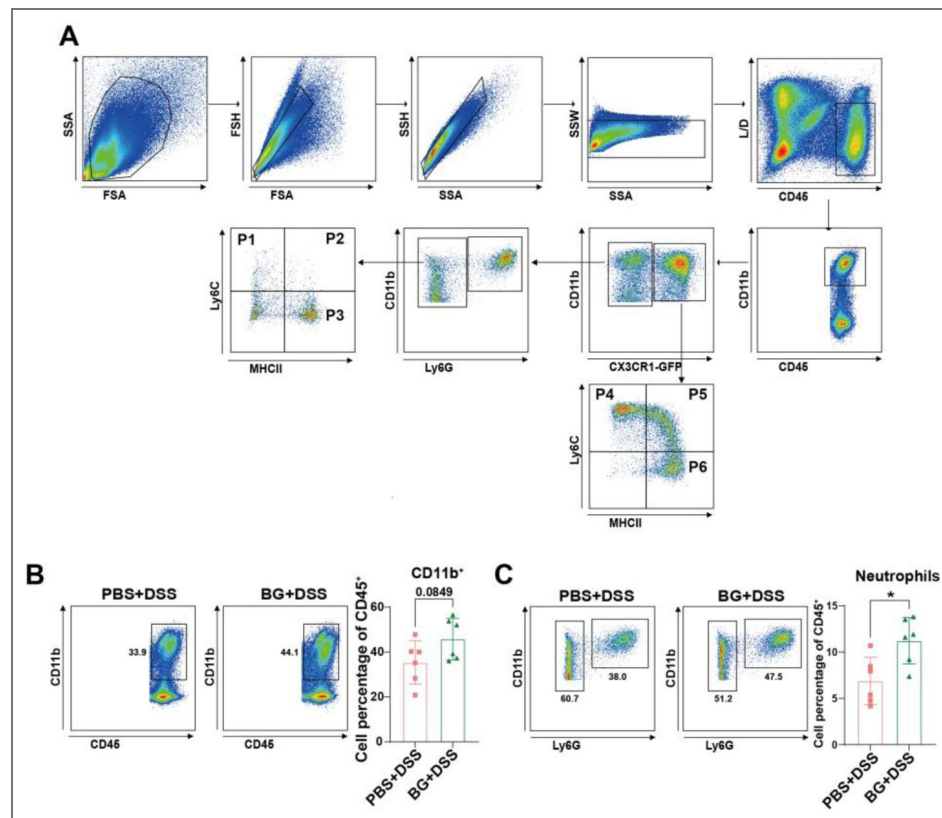


Fig S10. BG training upregulates the proportion of colonic monocytes/macrophages in colitis mice.

(A) Colonic LPMCs flow cytometry gating strategy, (B and C) The percentages of CD11b⁺ myeloid cells (B) and Ly6G⁺ neutrophils (C) were analyzed on day seven of colitis ($n = 6$). Data are presented as mean $\pm$ SD. Statistical significance: * $p < 0.05$. ns, not significant.

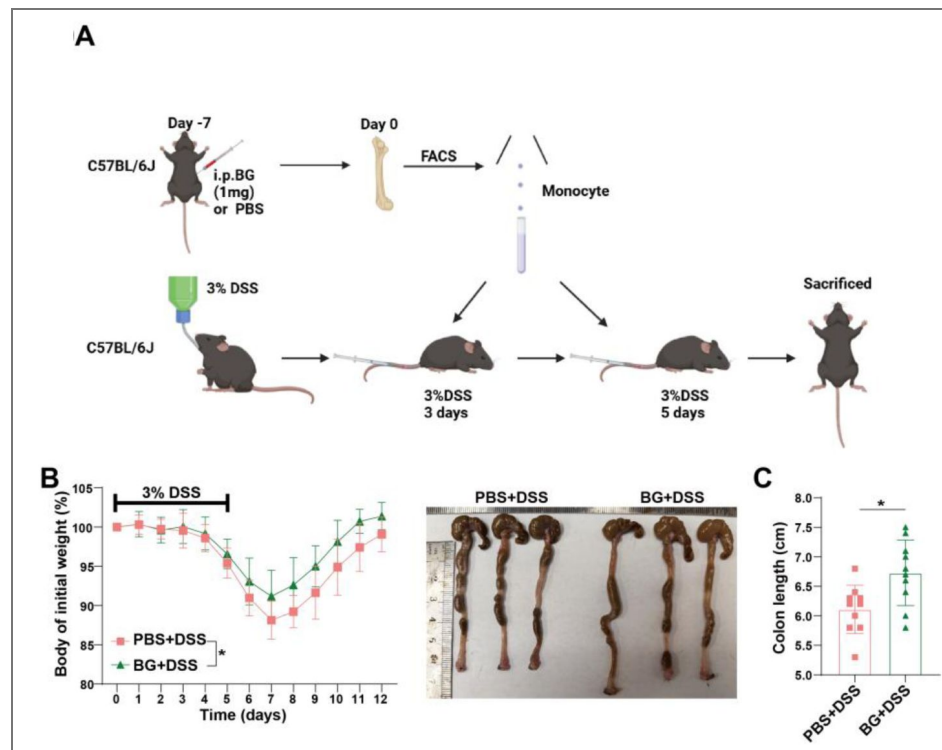


Fig S11. Adoptive transfer of BG-trained monocytes ameliorates experimental colitis.

(A) Bone marrow monocytes transplant and colitis treatment model, (B and C) Body weight change curve (B) and colon length changes (C) in WT colitis mice receiving Ly6C^{hi} monocyte adoptive transfer ($n = 10$). Data are presented as mean $\pm$ SD. Statistical significance: $*p < 0.05$.

Data availability

All data relevant to the study are included in the manuscript or uploaded as online supplemental information. Sequencing data have been deposited to the GEO with accession number GSE285859 (single cell RNA-seq), GSE285860 (RNA-seq).

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Additional information

Author contributions

Y.Y.L. designed and performed the experiments, analyzed the data and wrote the paper; Y.Y.F., Q.X.G. and Q.Y. C. analyzed the data, wrote the paper, and supervised the study. Y.Q.H., L.W., H.X.S., E.M.C., Q.Y.X., Y.C., Q.Q.F., L.L.L provided advice and oversaw a portion of the work. J.L.R., S.-C.C., H.Z.X. conceived the idea, secured funding, supervised the study, helped in data interpretations, revised drafts of manuscript; supervised overall project. All authors reviewed and approved the final version of the manuscript.

Ethics statements

Patient consent for publication

Not applicable.

Ethics approval

All animal protocols were reviewed and approved by the Institutional Animal Care and Use Committee of Xiamen University.

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Peer reviews

Reviewer #1 (Public review):

Summary:

This study presents an interesting investigation into the role of trained immunity in inflammatory bowel disease, demonstrating that β -glucan-induced reprogramming of innate immune cells can ameliorate experimental colitis. The findings are novel and clinically relevant, with potential implications for therapeutic strategies in IBD. The combination of functional assays, adoptive transfer experiments, and single-cell RNA sequencing provides

comprehensive mechanistic insights. However, some aspects of the study could benefit from further clarification to strengthen the conclusions.

Strengths:

- (1) This study elegantly connects trained immunity with IBD, demonstrating how β -glucan-induced innate immune reprogramming can mitigate chronic inflammation.
- (2) Adoptive transfer experiments robustly confirm the protective role of monocytes/macrophages in colitis resolution.
- (3) Single-cell RNA sequencing provides mechanistic depth, revealing the expansion of reparative Cx3cr1⁺ macrophages and their contribution to epithelial repair.
- (4) The work highlights the therapeutic potential of trained immunity in restoring gut homeostasis, offering new directions for IBD treatment.

Weaknesses:

While β -glucan may exert its training effect on hematopoietic stem cells, performing ATAC-seq on HSCs or monocytes to profile chromatin accessibility at antibacterial defense and mucosal repair-related genes would further validate the trained immunity mechanism. Alternatively, the authors could acknowledge this as a study limitation and future research direction.

Comments on revisions:

My concerns have been fully addressed. I have no additional comments.

<https://doi.org/10.7554/eLife.107339.2.sa3>

Reviewer #2 (Public review):

This study investigates how BG-induced myeloid reprogramming influences inflammatory bowel disease in a mouse model of DSS-induced colitis. The authors use in vivo functional experiments, adoptive transfer, and scRNA-seq to assess whether innate immune reprogramming can confer protection in colitis.

In the revised versions of the manuscript, the authors clarified the mechanistic scope of the study, softened the conclusions, and acknowledged the lack of direct epigenetic validation of trained immunity in this model. The manuscript now also better emphasizes the context-dependent nature of BG-induced reprogramming.

While some aspects remain correlative and will require further investigation, the central findings are well supported.

Overall, this work provides a meaningful contribution to the field, and I support its publication.

Comments on revisions:

No further comments.

<https://doi.org/10.7554/eLife.107339.2.sa2>

Reviewer #3 (Public review):

Summary:

In the present work, the authors offer evidence for the therapeutic potential of trained immunity in the context of inflammatory bowel disease (IBD). Prior research has demonstrated that innate cells pre-treated (trained) with β -glucan show an enhanced pro-inflammatory response upon a second challenge with the same or different stimulus. While an increased immune response can be beneficial and protect against bacterial infections, there is also the risk that it will worsen symptoms in various inflammatory disorders.

Remarkably, the authors show that β -glucan training of bone marrow hematopoietic progenitors and peripheral monocytes mitigates the pro-inflammatory effects of colitis, with protection extending to naïve recipients of the trained cells. Additionally, the authors demonstrate that mice preconditioned with β -glucan have enhanced resistance to *Staphylococcus aureus* and *Salmonella typhimurium* infections, indicating heightened immune responses.

Using a dextran sulfate sodium (DSS)-induced model of colitis, β -glucan pre-treatment significantly dampens disease severity. Importantly, the use of Rag1^{-/-} mice, which lack adaptive immune cells, confirms that the protective effects of β -glucan are mediated by innate immune mechanisms. Further, experiments using Ccr2^{-/-} mice underline the necessity of monocyte recruitment in mediating this protection, highlighting CCR2 as a key factor in the mobilization of β -glucan trained monocytes to inflamed tissues. In addition, β -glucan training highlights a distinct monocyte subpopulation with enhanced activation and phagocytic capacity. These monocytes, marked by increased expression of Cx3cr1, are suggestive of an increased ability to infiltrate inflamed colonic tissue and differentiate into macrophages.

Transcriptomic profiling reveals that β -glucan training upregulates genes associated with pattern recognition, antimicrobial defense, immunomodulation, and interferon signaling pathways, suggesting broad functional reprogramming of the innate immune compartment. Moreover, among the trained monocyte and macrophage subsets, gene expression signatures are associated with tissue and mucosal repair, suggesting a role in promoting resolution and regeneration following inflammatory insult. Furthermore, this was coupled with analysis of chromatin accessibility in publicly available data.

Strengths:

By employing a range of well-characterized murine models, the authors investigate specific mechanisms involved in the effects of β -glucan training. Furthermore, the study provides functional evidence that the protection conferred by the trained cells persists within the hematopoietic progenitors and can be transferred to naïve recipients. The integration of transcriptomic profiling allows the identification of changes in key genes and molecular pathways underlying the trained immune phenotype.

Weaknesses:

Further studies would benefit from investigating the cytokine responses of intestinal macrophages, particularly CX3CR1⁺ macrophages, following ex vivo stimulation of previously BCG-trained cells. Moreover, assessing the metabolic state of these macrophages would provide valuable insight into the mechanisms underlying trained immunity in this context.

Impact:

Overall, the authors present a mechanistically insightful investigation that advances our understanding of trained immunity in IBD. This is an important study that demonstrates that β -glucan-trained innate cells can confer protection against colitis and promote mucosal repair through trained-immunity related mechanisms. These findings underscore the potential of harnessing innate immune memory as a therapeutic approach for chronic inflammatory diseases.

<https://doi.org/10.7554/eLife.107339.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public review):

Summary:

This study presents an interesting investigation into the role of trained immunity in inflammatory bowel disease, demonstrating that β -glucan-induced reprogramming of innate immune cells can ameliorate experimental colitis. The findings are novel and clinically relevant, with potential implications for therapeutic strategies in IBD. The combination of functional assays, adoptive transfer experiments, and single-cell RNA sequencing provides comprehensive mechanistic insights. However, some aspects of the study could benefit from further clarification to strengthen the conclusions.

We are grateful for the reviewer's positive assessment of our study and constructive suggestions to improve the manuscript.

Strengths:

- (1) This study elegantly connects trained immunity with IBD, demonstrating how β -glucan-induced innate immune reprogramming can mitigate chronic inflammation.*
- (2) Adoptive transfer experiments robustly confirm the protective role of monocytes/macrophages in colitis resolution.*
- (3) Single-cell RNA sequencing provides mechanistic depth, revealing the expansion of reparative $Cx3cr1^+$ macrophages and their contribution to epithelial repair.*
- (4) The work highlights the therapeutic potential of trained immunity in restoring gut homeostasis, offering new directions for IBD treatment.*

Weaknesses:

While β -glucan may exert its training effect on hematopoietic stem cells, performing ATAC-seq on HSCs or monocytes to profile chromatin accessibility at antibacterial defense and mucosal repair-related genes would further validate the trained immunity mechanism. Alternatively, the authors could acknowledge this as a study limitation and future research direction.

We appreciate your comments on assessing the chromatin accessibility of HSCs induced by β -glucan training, as epigenetic reprogramming is known to be one of the underlying mechanisms for trained immunity suggested by many groups including our group. To delineate the genome-wide epigenetic reprogramming induced by β -glucan (BG), we reanalyzed publicly available chromatin profiling datasets where ATAC-seq of HSC from control and β -glucan trained mice was performed (accession number: CRA014389). Comparative analysis revealed HSC from BG-trained mice demonstrated pronounced enrichment at promoters and distal intergenic regions—key regulatory loci governing transcriptional activity (Fig. S7A). This divergent genomic targeting was further corroborated by distinct signal distribution profiles (Fig. S7B), supporting pronounced upregulation-driven remodeling of the epigenomic landscape induced by BG treatment. Functional annotation of these epigenetically primed promoters via GO term analysis revealed significant enrichment of immune-relevant processes, including leukocyte migration, cell-cell adhesion, and chemotaxis (Fig. S7C). Consistently, KEGG pathway analysis highlighted the enrichment of signaling cascades such

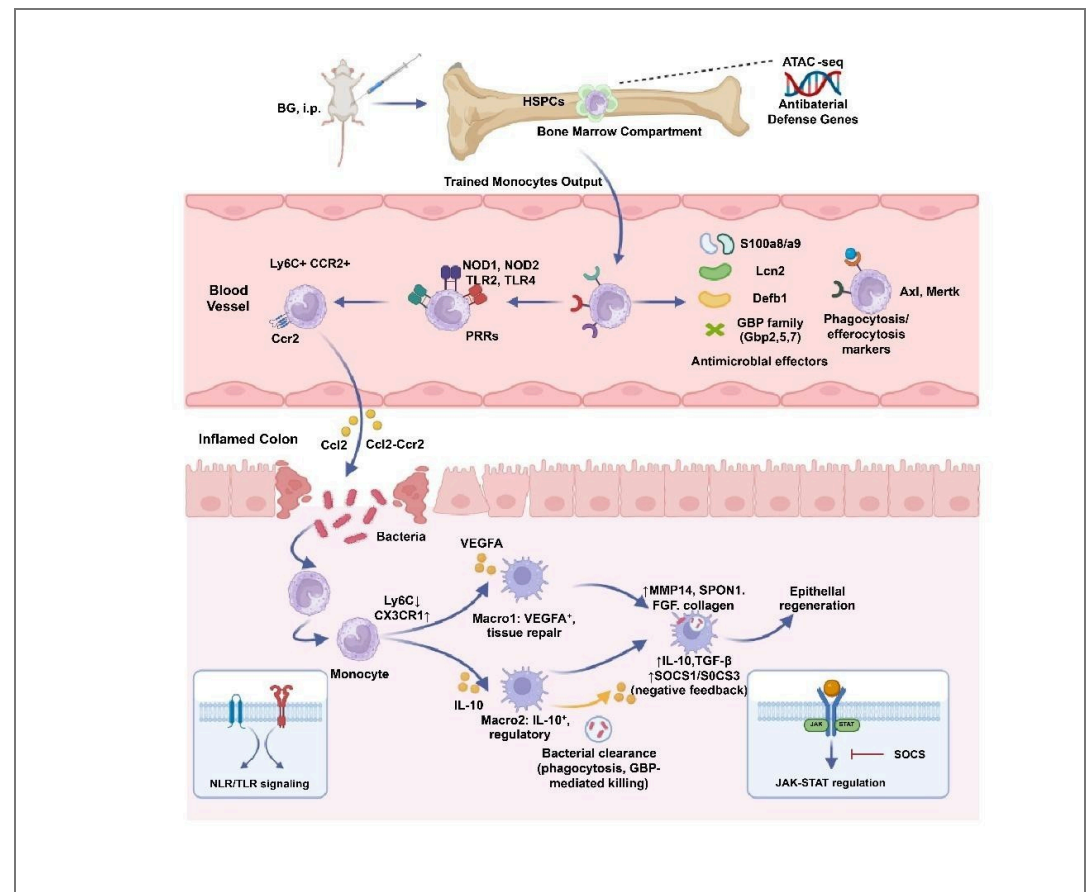
as chemokine signaling and cell adhesion molecules (Fig. S7D), reinforcing the involvement of BG-induced trained immunity in inflammatory and mucosal homing pathways.

Furthermore, promoter-centric enrichment of terms related to “defense response to bacterium” (Fig. S7E) underscored the role of BG in priming antibacterial transcriptional programs, which is a crucial axis for maintaining intestinal homeostasis. Locus-specific examination of chromatin states further validated BG-induced epigenetic modifications in the upstream regions of selected target genes, including Gbp5, Gbp2 and S100a8 and Nos2 (Fig. S7F). Collectively, our integrative reanalysis demonstrates that BG reshapes the epigenomic architecture at regulatory elements, thereby orchestrating immune gene expression programs directly relevant to IBD pathophysiology and mucosal immunity. (Line 201-211)

Reviewer 1 (Recommendations for the authors):

(1) It's better to include a schematic summarizing the proposed mechanism for reader clarity.

We appreciate your comments and proposed a graphical abstract as in Author response image 1.



Author response image 1.

(2) Discuss potential off-target effects of β -glucan-induced trained immunity (e.g., risk of exacerbated inflammation in other contexts).

We appreciate this important comment regarding the potential off-target or side-effects of β -glucan induced trained immunity. As trained immunity is known to augment inflammatory responses upon heterologous stimulation and has been implicated in chronic inflammation–

prone conditions such as atherosclerosis, this is an important consideration. Previous in vivo studies have shown that β -glucan pretreatment can enhance antibacterial or antitumor responses without inducing basal inflammation after one week of administration (PMID: 22901542, PMID: 30380404, PMID: 36604547, PMID: 33125892). Nevertheless, it remains possible that β -glucan-induced trained immunity could have unintended effects in certain contexts, which warrants further investigation and caution. We have discussed this potential caveat in the discussion (Lines 299-302)

Reviewer #2 (Public review):

Summary:

The study investigates whether β -glucan (BG) can reprogram the innate immune system to protect against intestinal inflammation. The authors show that mice pretreated with BG prior to DSS-induced colitis experience reduced colitis severity, including less weight loss, colon damage, improved gut repair, and lowered inflammation. These effects were independent of adaptive immunity and were linked to changes in monocyte function.

The authors show that the BG-trained monocytes not only help control inflammation but confer non-specific protection against experimental infections (Salmonella), suggesting the involvement of trained immunity (TI) mechanisms. Using single-cell RNA sequencing, they map the transcriptional changes in these cells and show enhanced differentiation of monocytes into reparative CX3CR1⁺ macrophages. Importantly, these protective effects were transferable to other mice via adoptive cell transfer and bone marrow transplantation, suggesting that the innate immune system had been reprogrammed at the level of stem/progenitor cells.

Overall, this study provides evidence that TI, often associated with heightened inflammatory programs, can also promote tissue repair and resolution of inflammation. Moreover, this BG-induced functional reprogramming can be further harnessed to treat chronic inflammatory disorders like IBD.

Strengths:

(1) The authors use advanced experimental approaches to explore the potential therapeutic use of myeloid reprogramming by β -glucan in IBD.

(2) The authors follow a data-to-function approach, integrating bulk and single-cell RNA sequencing with in vivo functional validation to support their conclusions.

(3) The study adds to the growing evidence that TI is not a singular pro-inflammatory program, but can adopt distinct functional states, including anti-inflammatory and reparative phenotypes, depending on the context.

We are grateful for your positive assessment of our study and recognition of its translational implications. We particularly appreciate the acknowledgment that our work expands the therapeutic potential of β -glucan-mediated trained immunity in ameliorating colitis.

Weaknesses:

(1) The epigenetic and metabolic basis of TI is not explored, which weakens the mechanistic claim of TI. This is especially relevant given that a novel reparative, antiinflammatory TI program is proposed.

We appreciate your valuable comment highlighting the importance of the epigenetic and metabolic basis of TI in providing mechanistic insight. While previous studies, including work from our group (S.-C. Cheng), have extensively characterized the epigenetic and metabolic signatures of monocytes from BG-trained mice—primarily in the context of

inflammatory genes—we acknowledge that these aspects are not directly addressed in our current manuscript as the current manuscript was aimed to build on the foundation of β -glucan-induced trained immunity established by many other groups including us and address its potential as a therapeutic approaches in the colitis setup.

That being said, we fully agree with your comments to analyze the epigenetic profile on key pathways similar to the question raised by reviewer 1, we reanalyze the relevant public datasets and presenting summarize the finding in Supplementary Figure S7. ATAC-seq analysis further validated and provide the epigenetic basis of the enhanced inflammatory and antibacterial capacity of monocytes which are seeded back in the HSC compartment.

(2) The absence of a BG-only group limits interpretation of the results. Since the authors report tissue-level effects such as enhanced mucosal repair and transcriptional shifts in intestinal macrophages (colonic RNA-Seq), it is important to rule out whether BG alone could influence the gut independently of DSS-induced inflammation. Without a BG-only control, it is hard to distinguish a true trained response from a potential modulation caused directly by BG.

We thank the reviewer for this important suggestion. Although we did not perform qPCR for mucosal repair genes in Figure S1C and Figure S1D, our colon RNA-seq analysis in Figure 5G included a BG-only control group (Colitis_d0). These results indicate that BG preconditioning alone does not alter baseline expression of colon mucosal repair genes, supporting the conclusion that the observed effects occur in the context of DSS-induced inflammation.

(3) Although monocyte transfer experiments show protection in colitis, the fate of the transferred cells is not described (e.g., homing or differentiation into Cx3cr1⁺ macrophage subsets). This weakens the link between specific monocyte subsets and the observed phenotype.

We thank the reviewer for this important point. We acknowledge that direct in vivo tracking of the adoptively transferred monocytes to confirm their homing to the colon and differentiation into specific macrophage subsets would strengthen the mechanistic link. However, due to technical limitations in reliably tracing the fate of transferred cells in our experimental setting, we were unable to provide this direct evidence. Instead, we present a strong correlative and functional evidence chain that supports the proposed model:

(a) Following BG pretreatment, we observed a significant decrease in circulating Ly6Chi monocytes specifically at the peak of colitis (day 7, Fig. 5D), concurrent with a marked increase in monocytes/macrophages within the colonic lamina propria (Fig. 2D). This inverse relationship strongly suggests enhanced recruitment of monocytes from the blood into the inflamed colon upon BG training.

(b) Using *CX3CR1-GFP* reporter mice, we found that BG pretreatment led to an increased proportion of colonic myeloid cells in an intermediate state (P5: Ly6C⁺MHCII⁺CX3CR1⁺, Fig. 5F). This population represents monocytes actively undergoing differentiation into intestinal macrophages, supporting the idea that BG accelerates the monocyte-to-macrophage transition in situ.

(c) Our scRNA-seq analysis independently revealed an expansion of monocyte-derived macrophage clusters (e.g., Macro1, Macro2) in BG-treated mice, which express canonical tissue macrophage markers (including *Cx3cr1*) and genes associated with tissue repair (e.g., *Vegfa*, Fig. 4A, 5H, 5I).

These data collectively indicate that BG-trained monocytes exhibit enhanced capacity for colonic recruitment and preferential differentiation toward reparative macrophage subsets, which aligns with the protective phenotype observed after adoptive transfer. We have explicitly noted the absence of direct fate-mapping data as a limitation in the revised

Discussion and agree that future studies employing advanced tracing techniques would be valuable to definitively establish this cellular trajectory. (Line 378-380)

(4) While scRNA-seq reveals distinct monocyte/macrophage subclusters (Mono1-3.), their specific functional roles remain speculative. The authors assign reparative or antimicrobial functions based on transcriptional signatures, but do not perform causal experiments (depletion or in vitro assays). The biological roles of these cells remain correlative.

We agree that the functional role of CX3CR1⁺ macrophages is not comprehensively validated and is currently inferred from scRNA-seq clustering. While our flow cytometry data show increased CX3CR1⁺ macrophages in the BG-TI group, and our CCR2 KO and monocyte adoptive transfer experiments indicate these macrophages are monocyte-derived, suggesting at least that β -glucan pretreatment alters the monocyte capacity which directly contribute to the enhanced colitis alleviation phenotype as observed. However, due to the fact that we fail to find a cluster dependent marker, which is also the current biggest caveats of the scRNAseq defined cell subclusters, we were not able to show direct casual evidence via specifically depleting subcluster cells. However, the result from the monocyte adoptive transfer experiment with Ccr2 KO mice experimental strongly suggest the presence of monocytes is crucial for this protective effect. We fully acknowledge this as a limitation of current study and clarify in the discussion that our conclusions regarding CX3CR1⁺ macrophage function are mainly based on transcriptional profiling and association with protective phenotypes, rather than direct causal evidence (Lines 400-404).

(5) While Rag1^{-/-} mice were used to rule out adaptive immunity, the potential role of innate lymphoid cells (ILCs), particularly ILC2s and ILC3s, which are known to promote mucosal repair (PMID: 27484190 IF: 7.6 Q1 IF: 7.6 Q1 IF: 7.6 Q1), was not explored. Given the reparative phenotype observed, the contribution of ILCs remains a confounding factor.

We appreciate your valuable comment regarding the potential role of ILCs in the observed mucosal repair. Indeed, in our current manuscript examining the BG-trained immunity effect, the contribution of ILCs was not evaluated. Due to the fact that adoptive transfer of trained monocytes into CCR2 KO mice could recapitulate the colitis alleviation phenotype, we think at least the β -glucan enhanced protection are dependent on trained monocytes. While acknowledge that the limitation and we could not rule out the possible role of ILCs in this process and discuss this limitation in the discussion in the revised manuscript.

The literature (PMID: 21502992; PMID: 32187516) supports a role for ILC3-mediated IL-22 production in tissue repair, which could overlap with our observed effects. However, our monocyte adoptive transfer experiments show that monocytes alone can alleviate DSS-induced colitis, suggesting a dominant role for monocytes in this context. Nonetheless, we will make it clear that ILC contributions cannot be excluded. (Line 322-326).

Reviewer 2 (Recommendations for the authors):

(1) The authors do not provide direct mechanistic evidence of TI (e.g., epigenetic and metabolic reprogramming). The absence of such data weakens the mechanistic strength of the TI claim. The authors should soften the terminology to BG-induced myeloid reprogramming suggestive of trained immunity, acknowledge, and discuss this limitation.

We appreciate your comment highlighting the lack of direct epigenetic and metabolic assessment in our current study. Previous work from our group (S.-C. Cheng) and others has extensively documented the epigenetic and metabolic profiles of monocytes from β -glucan-trained mice, focusing primarily on inflammatory-related genes. Based on this established

foundation, our current manuscript focuses on exploring the translational potential of BG-induced trained immunity.

That said, as mentioned in our response to the identified weakness, we performed reanalysis from the public epigenetic datasets with a focus on pathways related to reparative and antibacterial functions and integrated this part in the revised manuscript (Fig S7, Lines 201-211).

(2) CX3CR1⁺ macrophages' role is not functionally validated. The data relies solely on scRNA-seq and cluster annotations, which are insufficient to confirm functional roles in vivo. Depletion or in vitro studies would provide stronger causal evidence. The authors should acknowledge this limitation in the Discussion.

We agree that the functional role of CX3CR1⁺ macrophages is not comprehensively validated and is currently inferred from scRNA-seq clustering. While our flow cytometry data show increased CX3CR1⁺ macrophages in the BG-TI group, and our CCR2 KO and monocyte adoptive transfer experiments indicate these macrophages are monocyte-derived, suggesting at least that β -glucan pretreatment alters the monocyte capacity which directly contribute to the enhanced colitis alleviation phenotype as observed. However, due to the fact that we fail to find a cluster dependent marker, which is also the current biggest caveats of the scRNAseq defined cell subclusters, we were not able to show a direct casual evidence. We fully acknowledge this as a limitation of current study and clarify in the discussion that our conclusions regarding CX3CR1⁺ macrophage function are mainly based on transcriptional profiling and association with protective phenotypes, rather than direct causal evidence (Lines 395-404).

(3) Rag1^{-/-} mice retain innate lymphoid cells (ILCs), particularly ILC3, which are mucosal and produce IL-22, contributing to tissue repair (PMID: 21502992; PMID: 32187516). The potential for BG to activate ILCs remains unexplored in this study. This limits the interpretation of whether the observed protection arises from monocyte/macrophage reprogramming or is partially mediated by residual ILC activity. The authors should explicitly acknowledge this limitation and discuss the possible contribution of ILCs to the observed phenotype.

We appreciate your valuable comment regarding the potential role of ILCs in the observed mucosal repair. Indeed, in our current manuscript examining the BG-trained immunity effect, the contribution of ILCs was not evaluated. Due to the fact that adoptive transfer of trained monocytes into CCR2 KO mice could recapitulate the colitis alleviation phenotype, we think at least the β -glucan enhanced protection are dependent on trained monocytes. While acknowledge that the limitation and we could not rule out the possible role of ILCs in this process and discuss this limitation in the discussion in the revised manuscript

The literature (PMID: 21502992; PMID: 32187516) supports a role for ILC3-mediated IL-22 production in tissue repair, which could overlap with our observed effects. However, our monocyte adoptive transfer experiments show that monocytes alone can alleviate DSS-induced colitis, suggesting a dominant role for monocytes in this context. Nonetheless, we will make it clear that ILC contributions cannot be excluded. (Line 322-327).

(4) Figure 1-It would help to clarify whether a BG-only control group (without DSS) was included in the design. This would be critical to determine if BG alone alters the colon. If omitted, the authors should clearly state this and consider adding such a group in future experiments. This would help define the baseline effects of BG and support the claim that its benefits are dependent on TI (upon second challenge - DSS).

We appreciate this valuable suggestion. While we did not perform qPCR to assess mucosal repair genes in Figure S1C and Figure S1D, our colon RNA-seq analysis in Figure 5G included

a dedicated BG-only control group at based line before DSS treatment (Colitis_d0). These data indicate that BG preconditioning alone does not alter the baseline expression of colon mucosal repair genes.

(5) Figure 3 - It would strengthen the conclusions to include a vehicle-treated PBS BMT donor control group, or to state its absence. It is unclear whether the protective effect observed in recipients of BG-treated BM is due to trained immunity or to non-specific effects of transplantation, irradiation, or batch variation.

We fully agree with your comments that it is critical to including the vehicle-treated PBS BMT control to rule out any non-specific effects induced by transplantation, irradiation or batch variation. We actually did the blank PBS transfer control everytime after mice received irradiation treatment as a control to assess the successful induction of irradiation to get rid of bone marrow from irradiated mice. Mice that receive PBS only will die after 8 days while only mice receiving either bone marrow from PBS control or BG-treatment group will survive. We also perform flowcytometry to examine the successful BMT transplantation (Fig S5C). We have added part regarding the vehicle-treated control for BMT in the material method section for clarification (Lines 456-466).

(6) No gene expression or phenotypic data is provided for monocytes/macrophages in BMT recipients; therefore, it cannot be confidently stated that these cells were reprogrammed. Expression/phenotypic data should be added or discussed.

We thank the reviewer for raising this important point. We acknowledge that a detailed transcriptomic or phenotypic analysis of donor-derived tissue-resident myeloid cells in the BMT recipients would provide the most direct evidence for their reprogrammed state.

While our BMT study focused primarily on assessing the transferability of the protective phenotype via endpoint disease parameters and circulating immune cell composition, we present a coherent and compelling line of evidence supporting the conclusion that BG's training effect is maintained within the hematopoietic system of recipients and mediated by reprogrammed myeloid cells:

(a) A key finding is the significant increase in the proportion of donor-derived Ly6Chi monocytes in the peripheral blood of recipients receiving BG-trained bone marrow (Fig. 3J). This is not a bystander effect but direct evidence that the BG-induced on donor hematopoietic stem/progenitor cells instructs a biased differentiation program towards a specific effector precursor population within the new host, demonstrating the functional persistence of the trained state post-transplantation.

(b) The core of reprogramming in trained immunity lies in persistent epigenetic and functional changes. Our new analysis of public datasets (Fig. S7) confirms that BG directly reshapes the chromatin accessibility landscape in hematopoietic stem cells (HSCs), particularly at loci regulating immune and antibacterial responses. This provides the fundamental mechanism explaining how the trained phenotype is both long-lasting and transplantable: the reprogramming occurs at the progenitor level.

(c) The most causally compelling data in our study comes from the independent adoptive transfer experiment, where transfer of purified BG-trained monocytes alone was sufficient to ameliorate colitis in recipient mice (Fig. 3K, L). This definitively proves that the trained monocytes themselves carry the protective functional program. It strongly suggests that these reprogrammed monocytes/macrophages are the likely effectors mediating protection in the BMT model.

(d) Our interpretation aligns with well-established paradigms in the field. Precedent studies confirm that the BG-trained phenotype (e.g., enhanced cytokine potential) can be transferred via BMT or monocyte adoption. For instance, Haacke et al. (PMID: 40020679) demonstrated

that splenic monocytes from BG-trained donors, when transferred into arthritic recipient mice, led to elevated inflammatory cytokine (e.g., *Tnf*, *Il6*) expression in recipient joints, directly proving the maintained functional reprogramming of trained cells in a heterologous host environment. This provides a strong precedent supporting the functional activity of transferred trained cells in our model.

(7) The study is consistent with emerging evidence that distinct TI programs may exist depending on the stimulus and context, including immunoregulatory and tissue-reparative responses (PMID: 35133977; PMID: 31732931; PMID: 32716363; PMID: 30555483). The authors should integrate this perspective into the Discussion to acknowledge that their findings may represent one example of such context-dependent, potentially reparative TI programs. This would place the study within the growing literature describing functional heterogeneity in innate immune training.

We appreciate this suggestion and have incorporated it into the discussion. In the revised manuscript, we discussed how our findings of BG-induced protective myeloid reprogramming align with the concept of tissue-reparative or immunoregulatory TI, which is distinct from the pro-inflammatory TI phenotypes described in other contexts. By highlighting the functional heterogeneity of innate immune training, we position our work as an example of a stimulus-specific, reparative TI program. (Lines 356-379)

Reviewer #3 (Public review):

Summary:

In the present work, Yinyin Lv et al offer evidence for the therapeutic potential of trained immunity in the context of inflammatory bowel disease (IBD). Prior research has demonstrated that innate cells pre-treated (trained) with β -glucan show an enhanced pro-inflammatory response upon a second challenge.

*While an increased immune response can be beneficial and protect against bacterial infections, there is also the risk that it will worsen symptoms in various inflammatory disorders. In the present study, the authors show that mice preconditioned with β -glucan have enhanced resistance to *Staphylococcus aureus* infection, indicating heightened immune responses.*

The authors demonstrate that β -glucan training of bone marrow hematopoietic progenitors and peripheral monocytes mitigates the pro-inflammatory effects of colitis, with protection extending to naïve recipients of the trained cells.

*Using a dextran sulfate sodium (DSS)-induced model of colitis, β -glucan pre-treatment significantly dampens disease severity. Importantly, the use of *Rag1*^{-/-} mice, which lack adaptive immune cells, confirms that the protective effects of β -glucan are mediated by innate immune mechanisms. Further, experiments using *Ccr2*^{-/-} mice underline the necessity of monocyte recruitment in mediating this protection, highlighting CCR2 as a key factor in the mobilization of β -glucan-trained monocytes to inflamed tissues. Transcriptomic profiling reveals that β -glucan training upregulates genes associated with pattern recognition, antimicrobial defense, immunomodulation, and interferon signaling pathways, suggesting broad functional reprogramming of the innate immune compartment. In addition, β -glucan training induces a distinct monocyte subpopulation with enhanced activation and phagocytic capacity. These monocytes exhibit an increased ability to infiltrate inflamed colonic tissue and differentiate into macrophages, marked by increased expression of *Cx3cr1*. Moreover, among these trained monocyte and macrophage subsets, other gene expression signatures are associated with tissue and mucosal repair, suggesting a role in promoting resolution and regeneration following inflammatory insult.*

Strengths:

(1) Overall, the authors present a mechanistically insightful investigation that advances our understanding of trained immunity in IBD.

(2) By employing a range of well-characterized murine models, the authors investigate specific mechanisms involved in the effects of β -glucan training.

(3) Furthermore, the study provides functional evidence that the protection conferred by the trained cells persists within the hematopoietic progenitors and can be transferred to naïve recipients. The integration of transcriptomic profiling allows the identification of changes in key genes and molecular pathways underlying the trained immune phenotype.

(4) This is an important study that demonstrates that β -glucan-trained innate cells confer protection against colitis and promote mucosal repair, and these findings underscore the potential of harnessing innate immune memory as a therapeutic approach for chronic inflammatory diseases.

Thank you for the positive evaluation and constructive feedback on our manuscript.

Weaknesses:

However, FPKM is not ideal for between-sample comparisons due to its within-sample normalization approach. Best practices recommend using raw counts (with DESeq2) for more robust statistical inference.

We appreciate the reminder about best practices for RNA-seq analysis. We apologize for the inaccurate description in the Materials and Methods section. For all differential expression analyses, we have in fact used raw count data as input for DESeq2. FPKM values were only used for visualization purposes, such as in heatmaps and clustering analyses. We correct this description in the revised manuscript to accurately reflect our analysis workflow. (Lines 488-499)

Reviewer 3 (Recommendations for the authors):

(1) Current best practices recommend working with raw count data when using DESeq2 to ensure statistically robust differential expression analysis between samples. However, for visualization and clustering, like heatmaps, FPKMs can be used. Could the authors explain why they have used FPKM for differential gene expression analysis?

We appreciate the reminder about best practices for RNA-seq analysis. We apologize for the inaccurate description in the Materials and Methods section. For all differential expression analyses, we have in fact used raw count data as input for DESeq2. FPKM values were only used for visualization purposes, such as in heatmaps and clustering analyses. We correct this description in the revised manuscript to accurately reflect our analysis workflow. (Lines 488-499)

Minor Comment

(1) Line 92: remove extra word "that".

We remove the extra word "that" from Line 92 in the revised manuscript.

(2) Line 201: please state here what "GBP" stands for, as it appears first.

We define "GBP" as "Guanylate-Binding Protein" at its first appearance in Line 201. (Lines 213)

(3) Line 235: consider rewriting "we analyzed the day 7 RNA-seq data, which revealed significant enrichment of the myeloid"; added spacing for "day 7", "which", and "the".

We revise the sentence in Line 235 to read: "We analyzed the day 7 RNA-seq data, which revealed significant enrichment of the myeloid..." to improve readability. (Lines

246-247)

(4) Line 290: consider rewriting "as seen in conditions such as rheumatoid arthritis and ...".

We revise Line 290 to: "as observed in conditions such as rheumatoid arthritis and..." for clarity. (Lines 301-302)

(5) Line 375-376: please check sentence starting lower case "with minor modifications, by assessing".

We correct the sentence to start with a capital letter: "With minor modifications, by assessing..." (Lines 422-423)

(6) Line 399: kindly consider adding "was" after "cDNA".

We revise Line 399 to include "was" as suggested: "cDNA was synthesized..." (Lines 446)

(7) Line 346-347: consider adding "which" after "monocytes": "We transferred BGpreconditioned monocytes which significantly alleviated clinical symptoms".

We revise Line 346-347 to include "which" as suggested for grammatical clarity. (Lines 385-386)

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